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MSSC2025
TORINO 8-12 SEP 2025

Large Scale DFT Simulations on CPU/GPU Architectures

Giacomo Ambrogio

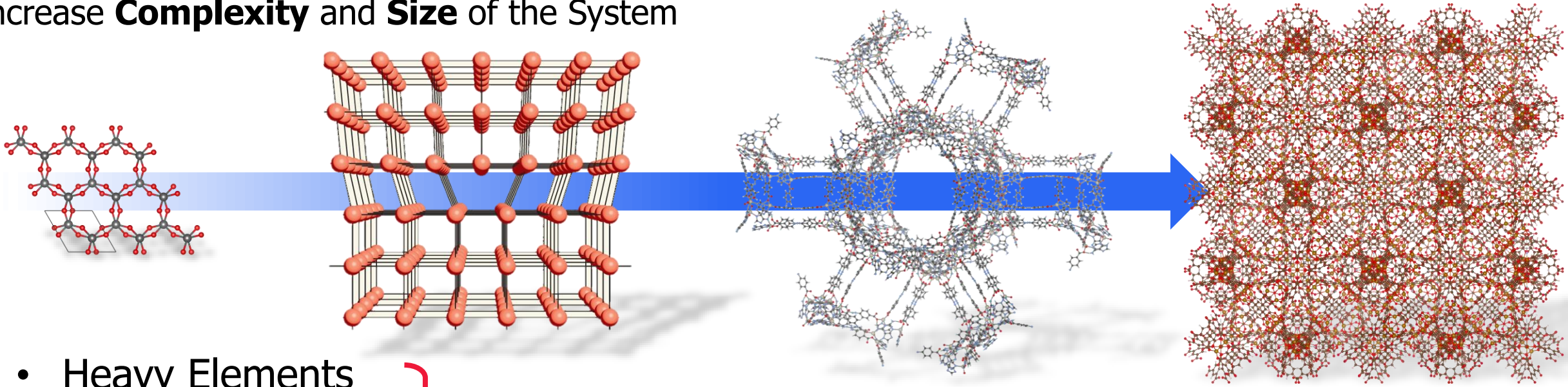
PhD Student, University Of Turin (Italy)

September 11th 2025
Molecular Biotechnology
Center

Ab Initio Modelling in Solid State Chemistry

Large Scale

Increase **Complexity** and **Size** of the System



- Heavy Elements
- SOC
- MOF
- Defects
- Moiré Patterns
- Phonon
- Etc.

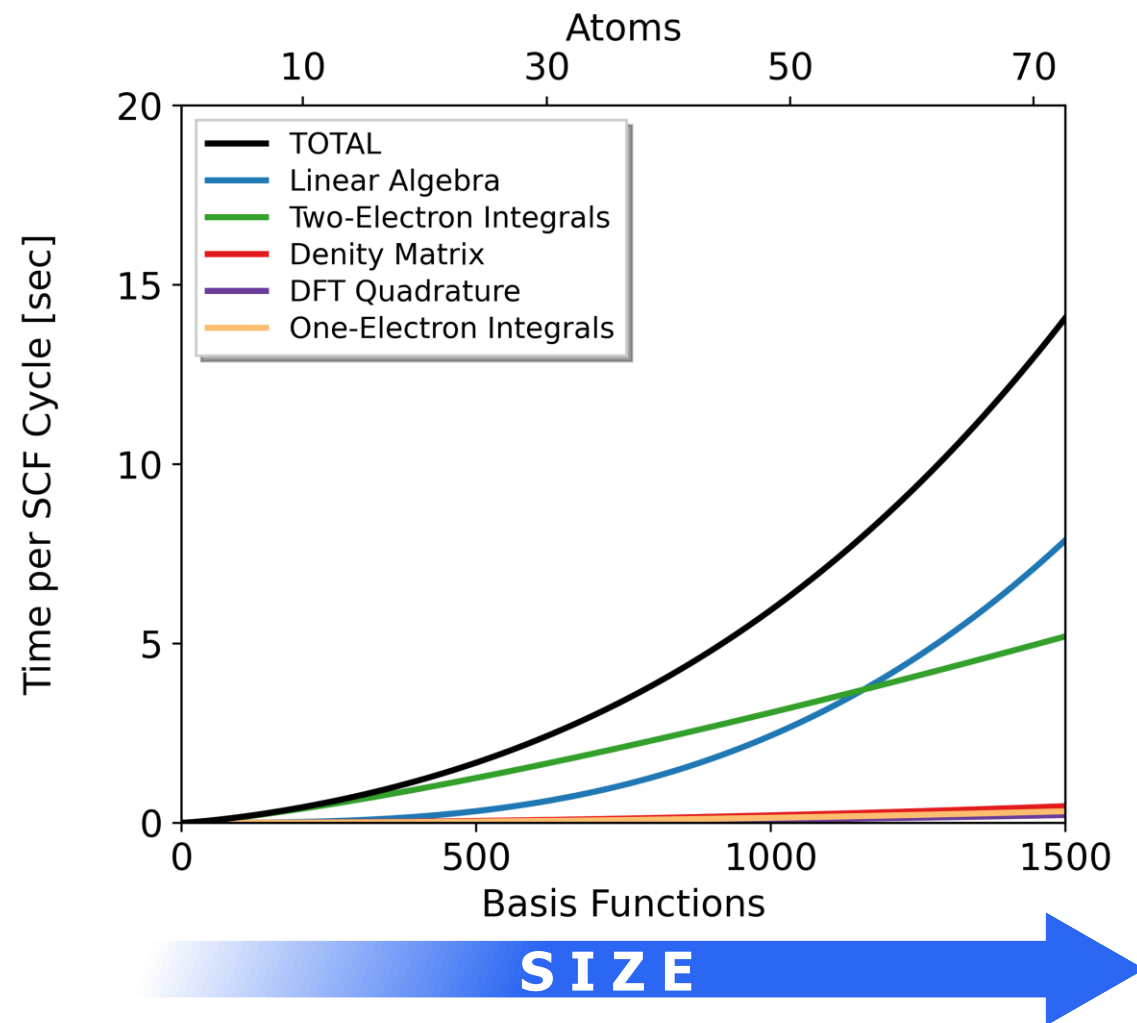
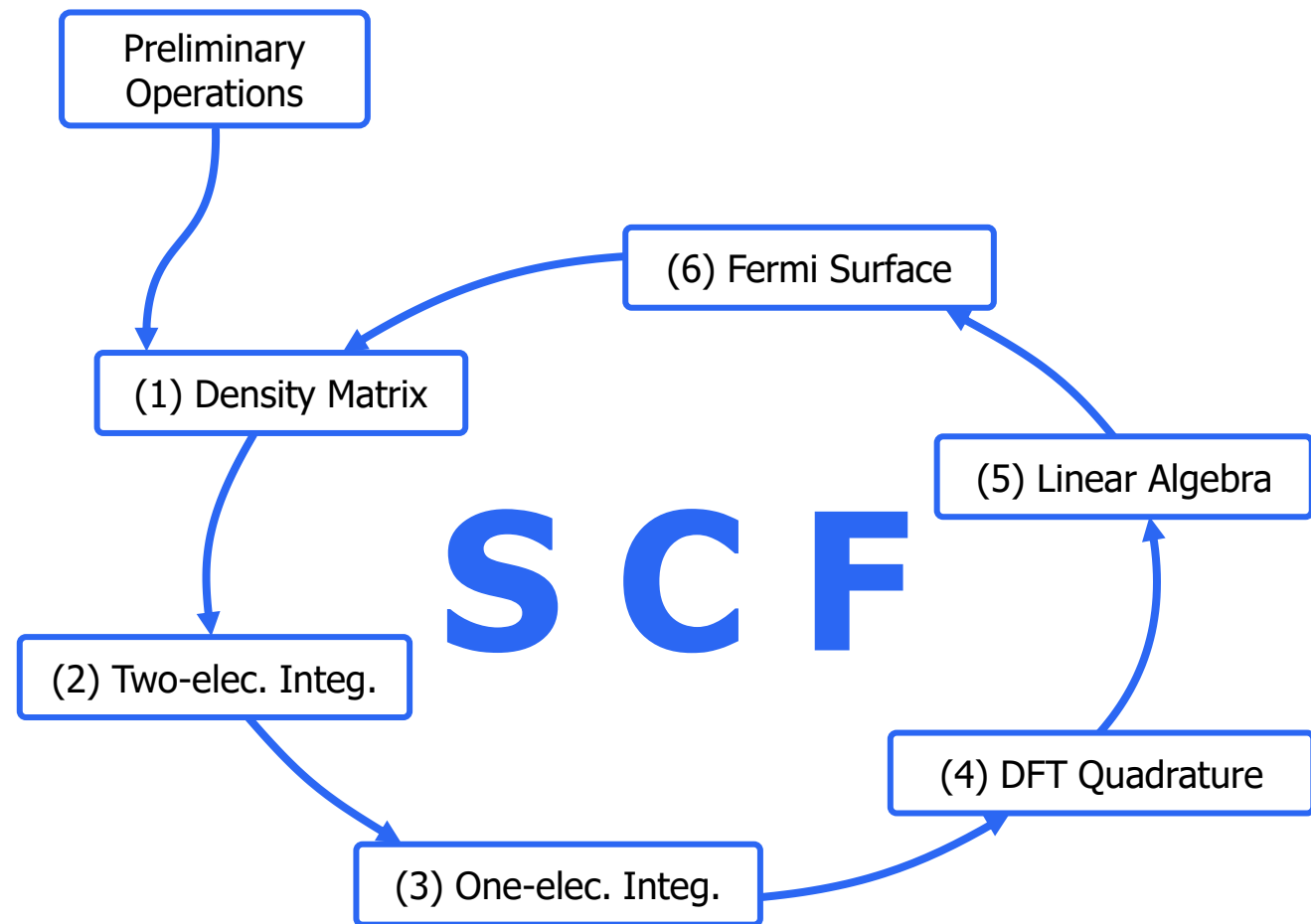
EXPENSIVE

Scalable Methods (DFT)

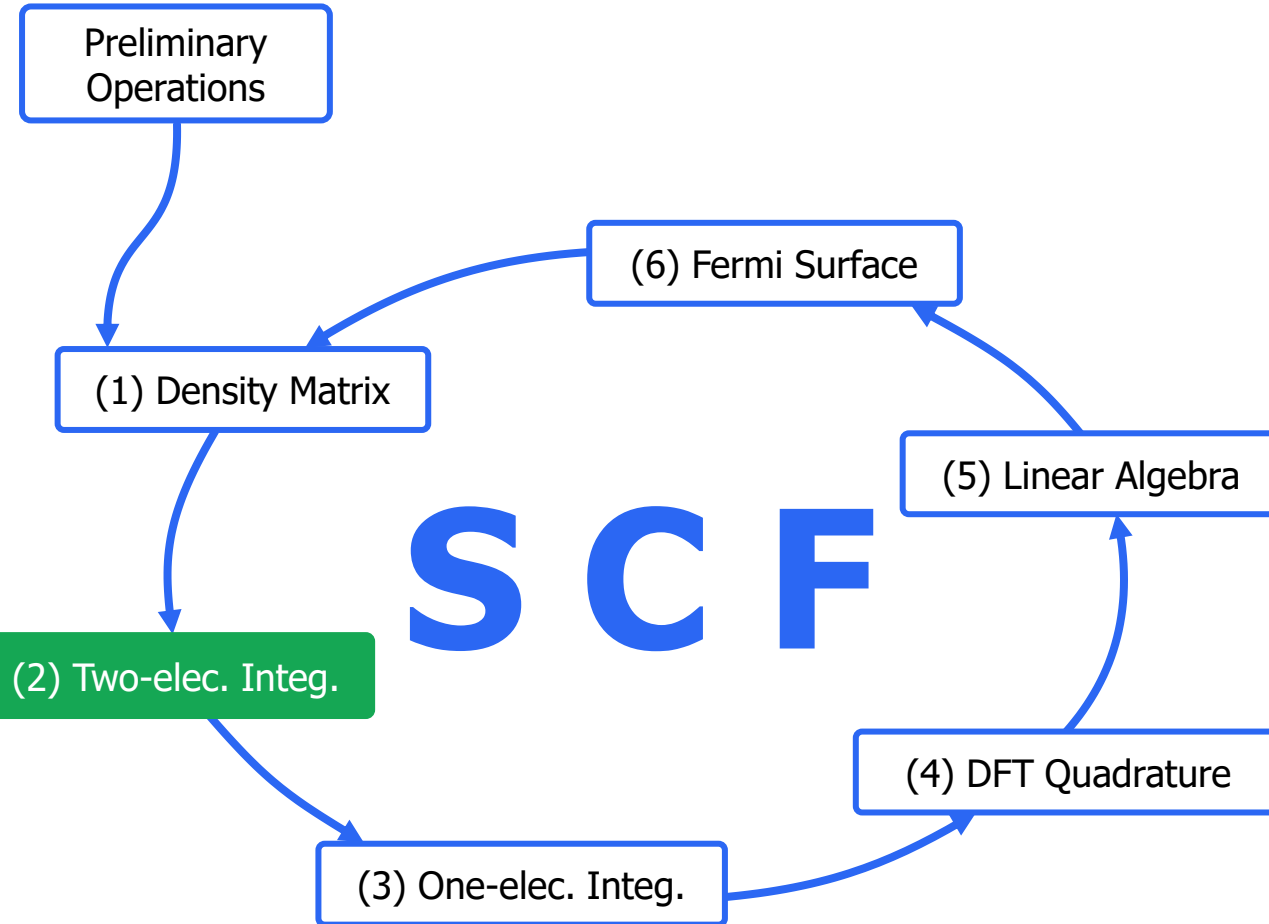
Improve the Code

Powerful Hardware

SCF: Code Efficiency



SCF: Two-electron Integrals



Parallel Computing



MPI
OpenMP

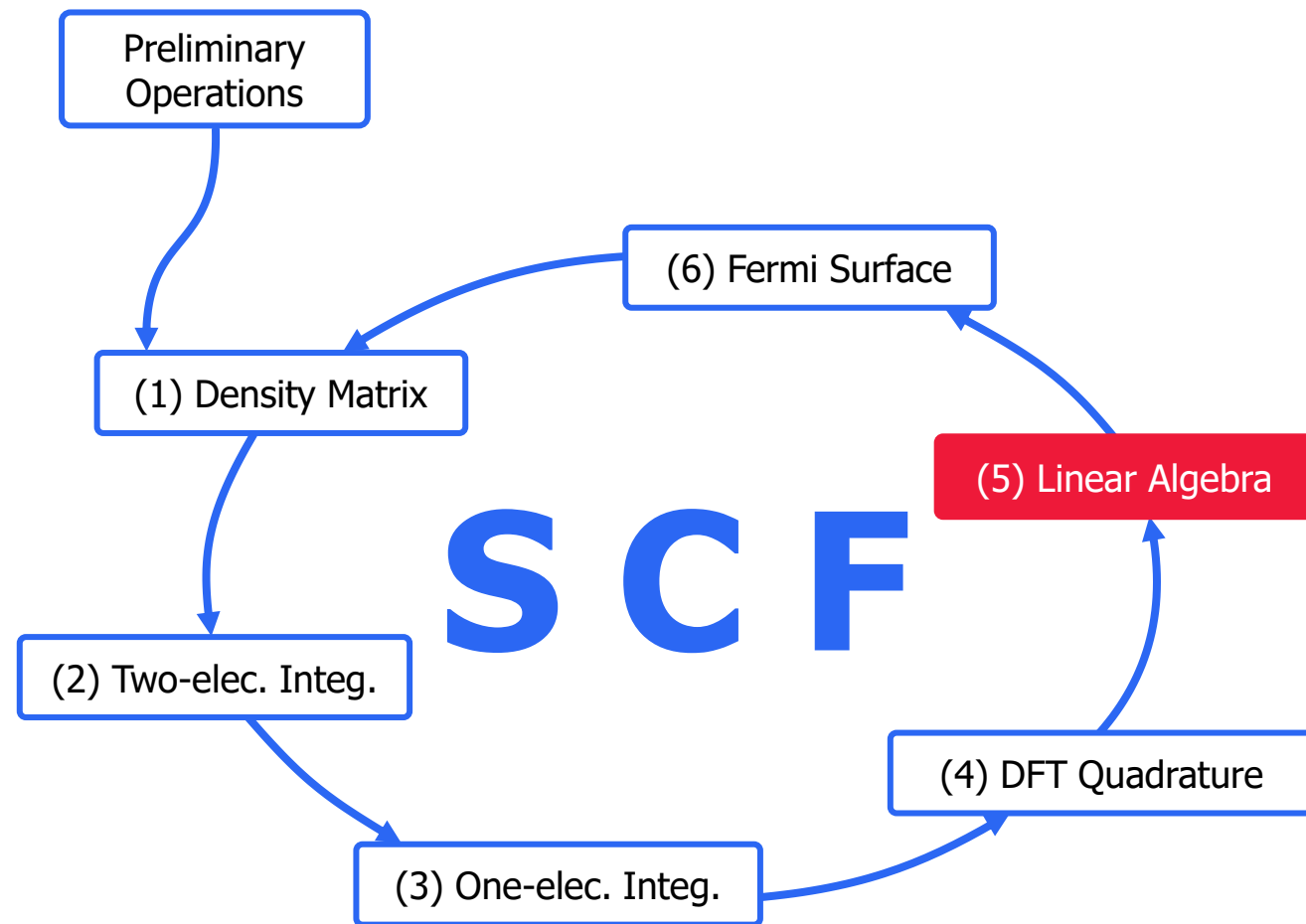
Coulomb

$$C_{\mu\nu}^{\mathbf{g}} = \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\nu}^{\mathbf{g}} | \chi_{\sigma}^{\mathbf{s}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Exact Exchange (Only Hybrid DFT and HF)

$$V_{\text{exx } \mu\nu}^{\mathbf{g}} = \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

SCF: Linear Algebra



SCF: Linear Algebra

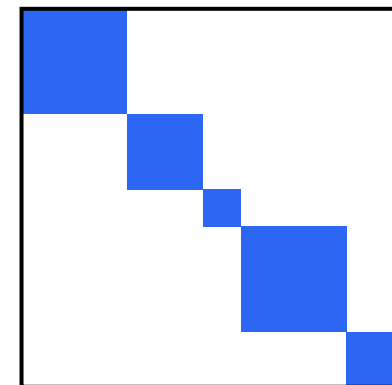
PCRYSTAL

Exploit Symmetry:

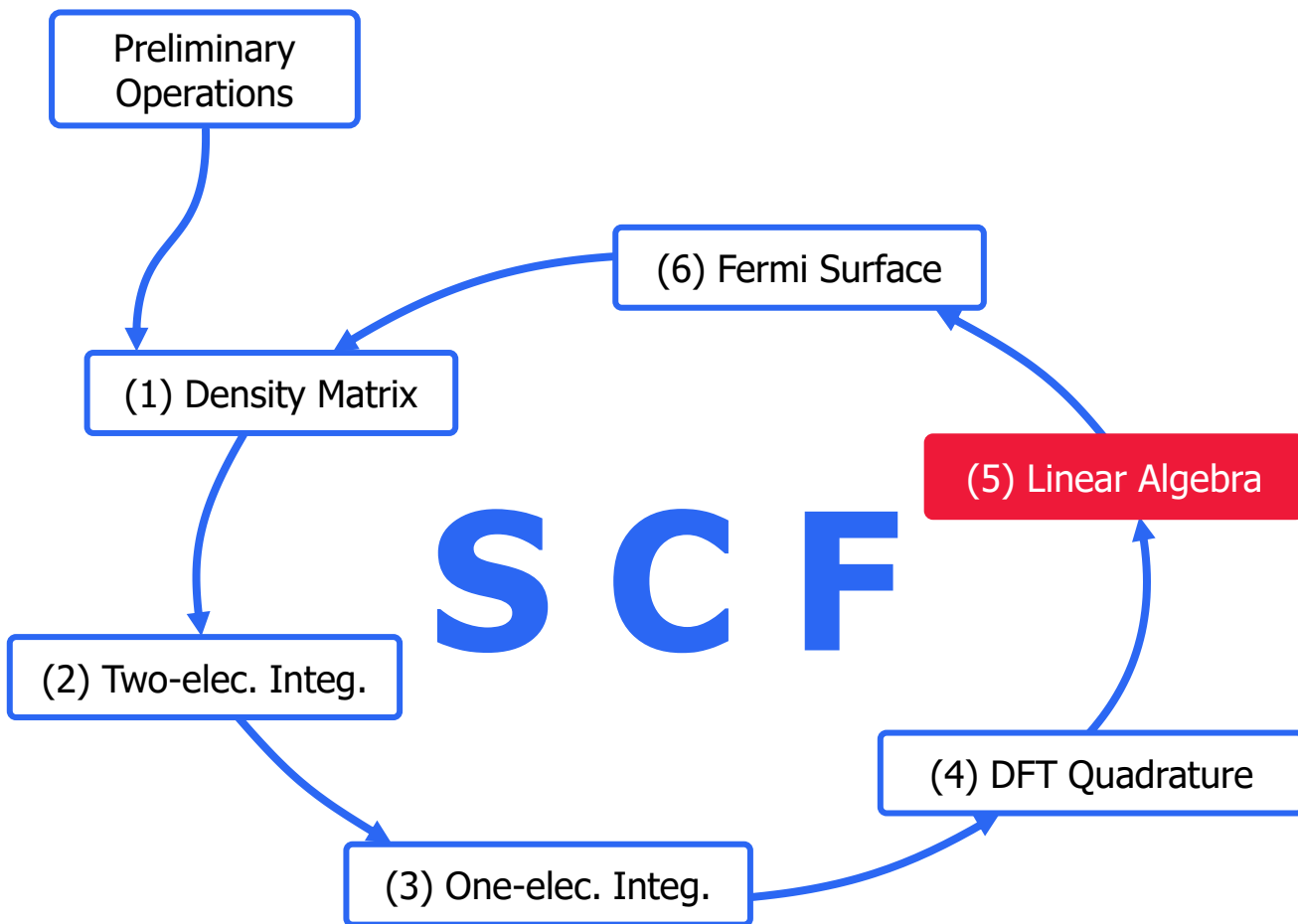
One matrix $\longrightarrow$ Block diagonal form

Each block is *independent*

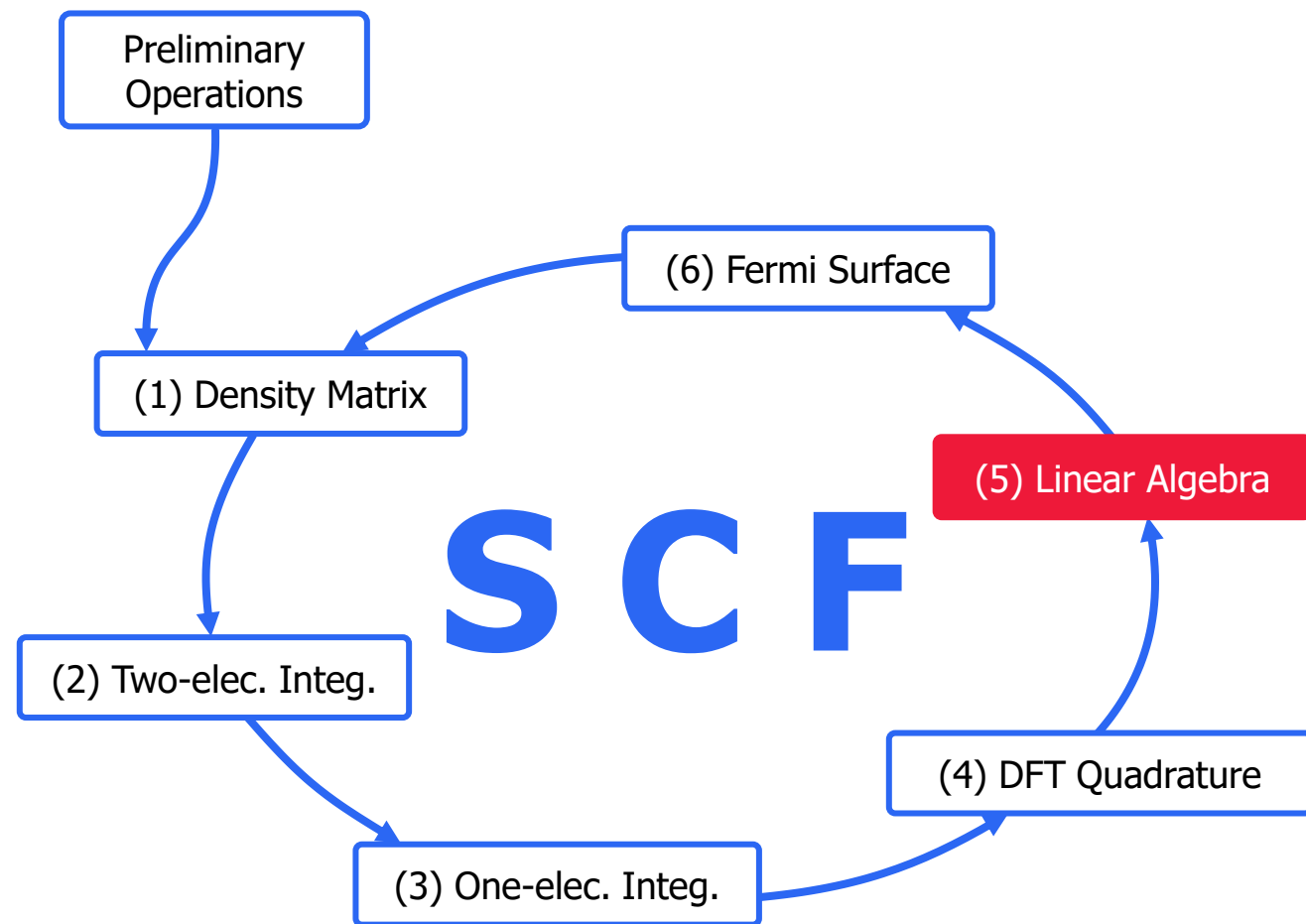
Assigned to a different CPU core



SCF



SCF: Linear Algebra



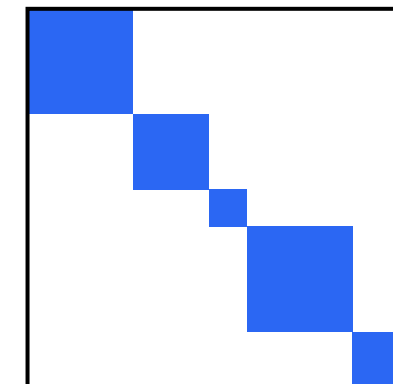
PCRYSTAL

Exploit Symmetry:

One matrix $\longrightarrow$ Block diagonal form

Each block is *independent*

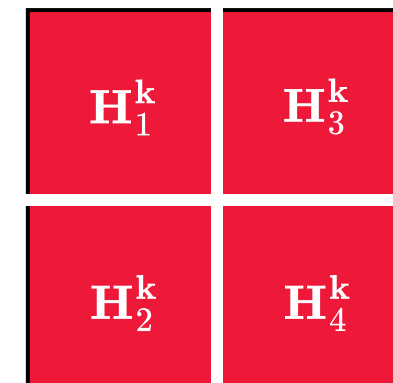
Assigned to a different CPU core



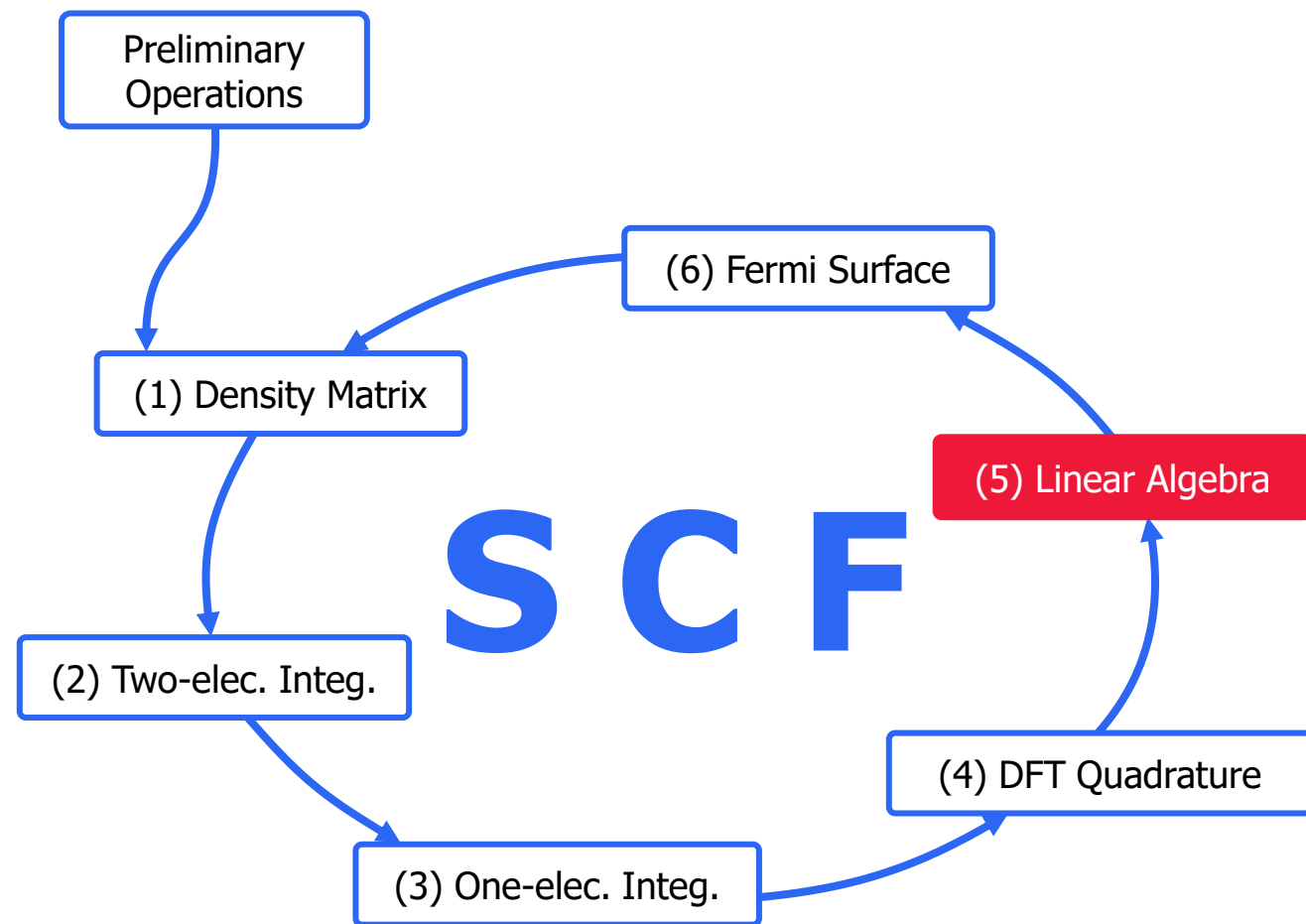
MPPCRYSTAL

Exploit ScaLAPACK:

LA library to perform *large diagonalization* on multiple CPU cores



SCF: Linear Algebra



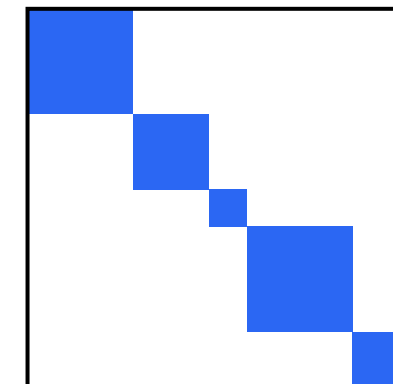
PCRYSTAL

Exploit Symmetry:

One matrix $\longrightarrow$ Block diagonal form

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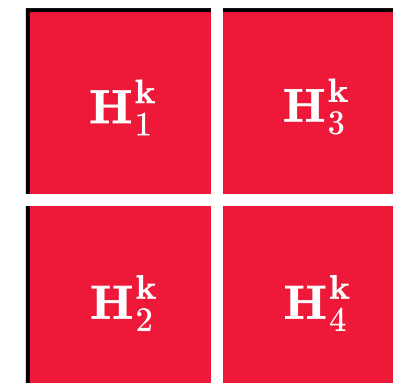
Assigned to a different CPU core



MPPCRYSTAL

Exploit ScaLAPACK:

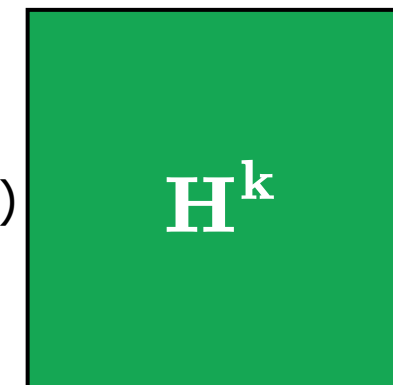
LA library to perform *large diagonalization* on multiple CPU cores



GCRYSTAL

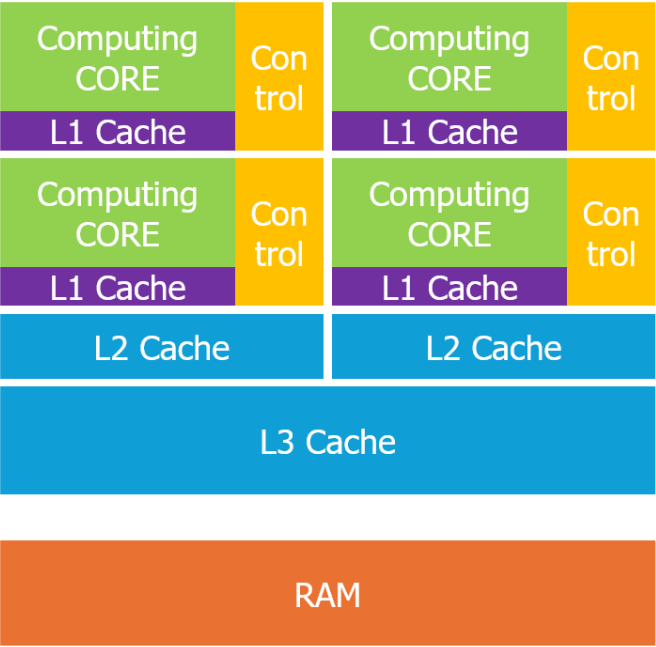
Exploit GPU:

Extremely good for *fast* (and big) LA operations



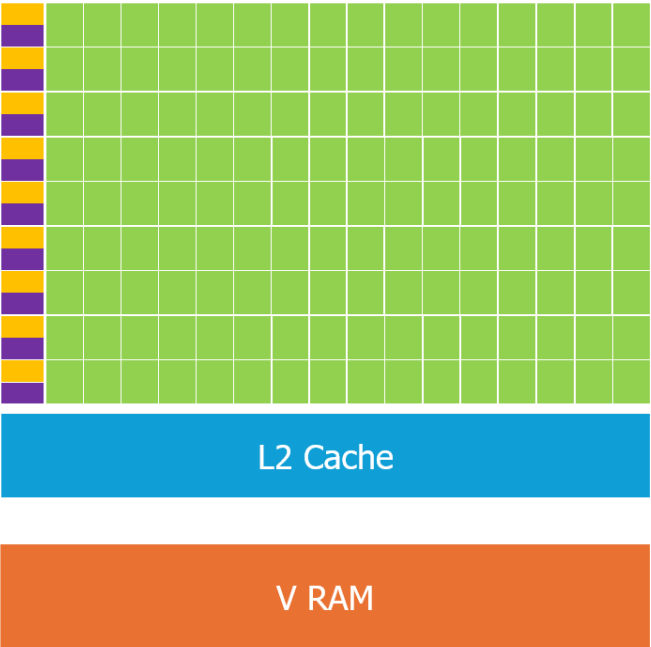
Hardware Architecture

CPU



- Minimize **Latency**
- Total independency of cores
- Powerful cores, **complex** operations

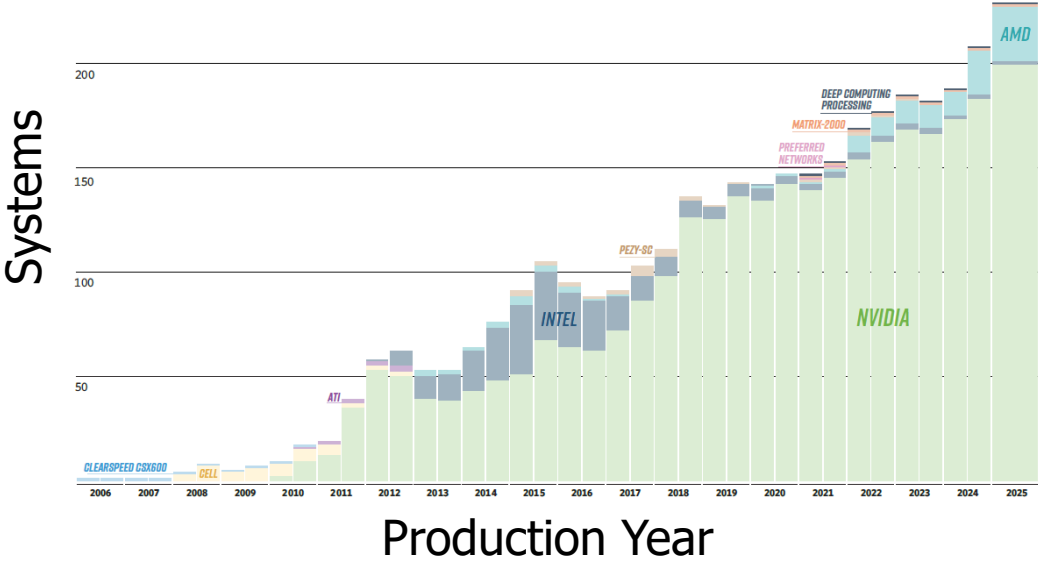
GPU



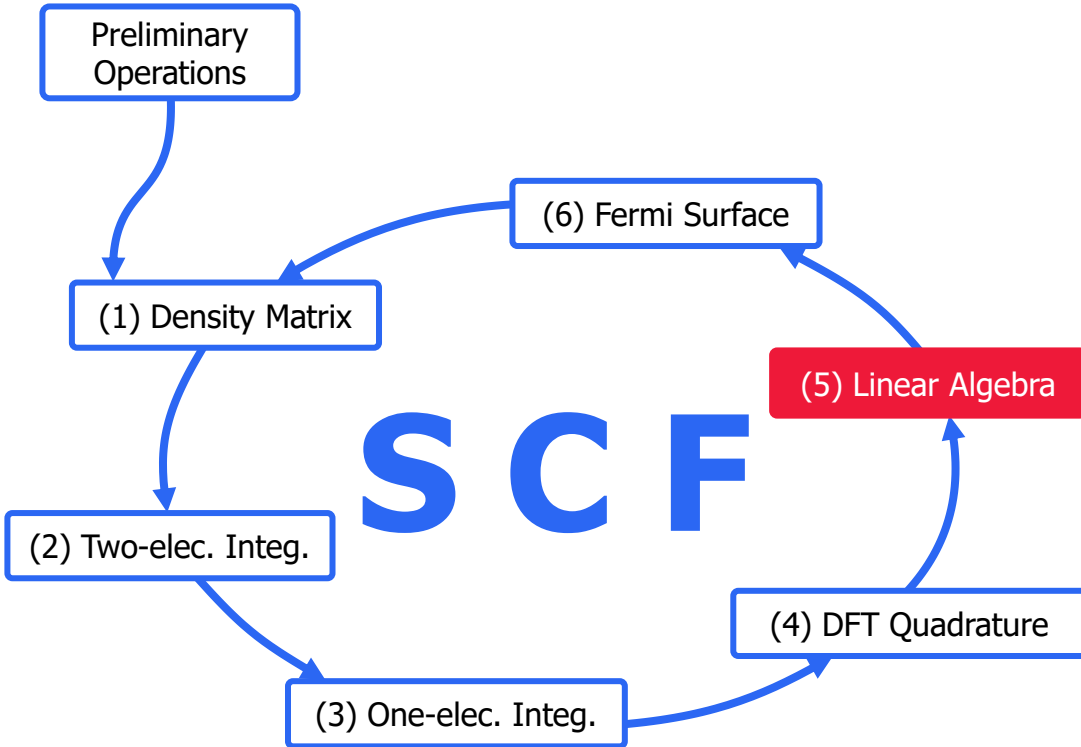
- Maximize **Throughput**
- Cores divided in blocks
- Small cores, **simple** operations
- Need dedicated Programming



ACCELERATORS/CO-PROCESSORS



Linear Algebra Step



$$\mathbf{H}^k \mathbf{A}^k = \mathbf{S}^k \mathbf{A}^k \mathbf{E}^k$$

Fourier Transform

$$H_{\mu\nu}^k = \sum_{\mathbf{g}} H_{\mu\nu}^{\mathbf{g}} e^{i\mathbf{k} \cdot \mathbf{g}}$$



Similarity Transformation

$$\tilde{\mathbf{H}}^k = \mathbf{T}^{k\dagger} \mathbf{H}^k \mathbf{T}^k$$

Diagonalization

$$\tilde{\mathbf{H}}^k \tilde{\mathbf{A}}^k = \tilde{\mathbf{A}}^k \epsilon^k$$

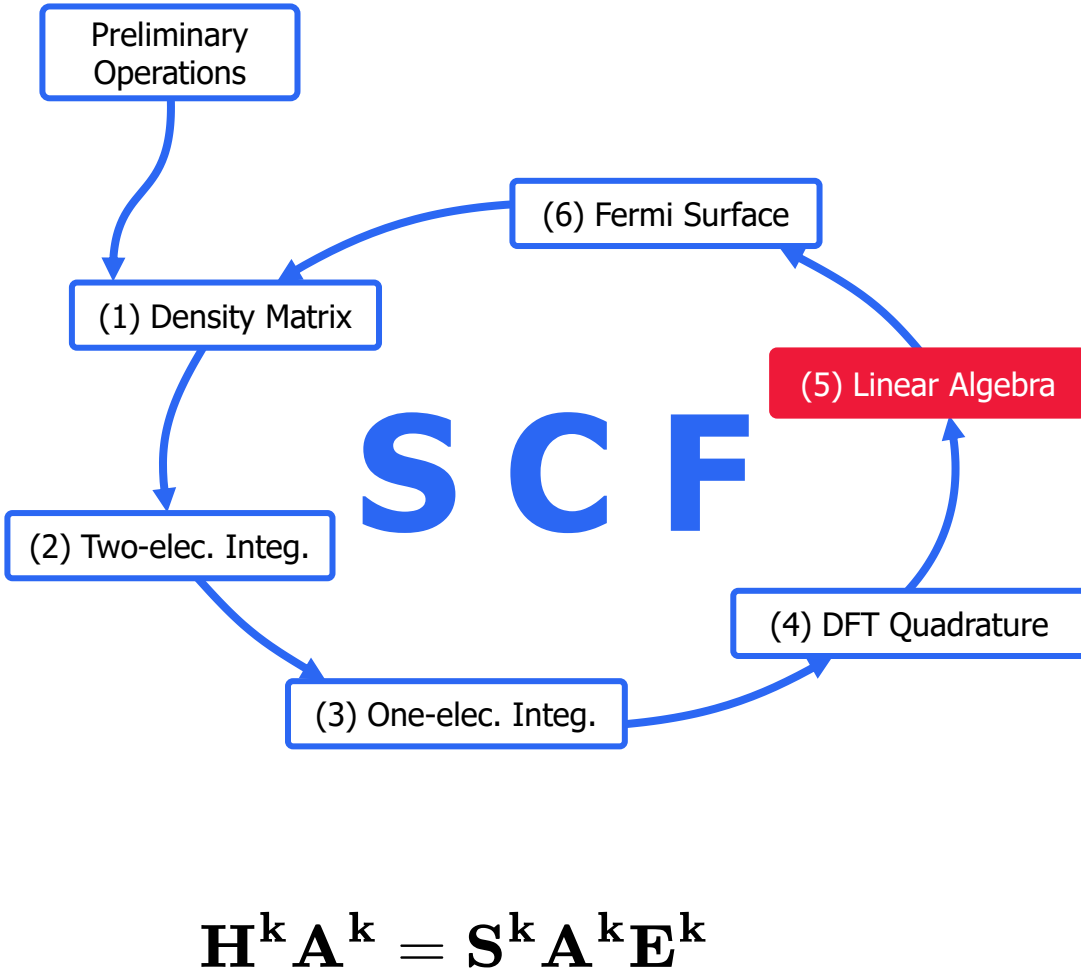
Energy Levels

Back-transformation

$$\mathbf{A}^k = \mathbf{T}^k \tilde{\mathbf{A}}^k$$

AO Coefficients

Linear Algebra Step



Fourier Transform

$$H_{\mu\nu}^k = \sum_{\mathbf{g}} H_{\mu\nu}^{\mathbf{g}} e^{i\mathbf{k} \cdot \mathbf{g}}$$



Similarity Transformation

$$\tilde{\mathbf{H}}^k = \mathbf{T}^{k\dagger} \mathbf{H}^k \mathbf{T}^k$$

Diagonalization

$$\tilde{\mathbf{H}}^k \tilde{\mathbf{A}}^k = \tilde{\mathbf{A}}^k \epsilon^k$$

Energy Levels

Back-transformation

$$\mathbf{A}^k = \mathbf{T}^k \tilde{\mathbf{A}}^k$$

AO Coefficients

GPU

MPI process 1

Loop *on assigned $\mathbf{k}$ points*

FT

$\mathbf{H}^{\mathbf{k}}$ $\longrightarrow$ GPU

$\mathbf{T}^{\mathbf{k}}$ $\longrightarrow$ GPU

Similarity

Diagonalization

Back-transform

$\mathbf{e}^{\mathbf{k}}$ $\longrightarrow$ CPU

$\mathbf{A}^{\mathbf{k}}$ $\longrightarrow$ CPU

End Loop

GPU 1

MPI process 2

Loop *on assigned $\mathbf{k}$ points*

FT

$\mathbf{H}^{\mathbf{k}}$ $\longrightarrow$ GPU

$\mathbf{T}^{\mathbf{k}}$ $\longrightarrow$ GPU

Similarity

Diagonalization

Back-transform

$\mathbf{e}^{\mathbf{k}}$ $\longrightarrow$ CPU

$\mathbf{A}^{\mathbf{k}}$ $\longrightarrow$ CPU

End Loop

GPU 2

GCRYSTAL is based on PCRYSTAL:
one *FULL* matrix for each process

Fourier Transform

$$H_{\mu\nu}^{\mathbf{k}} = \sum_{\mathbf{g}} H_{\mu\nu}^{\mathbf{g}} e^{i\mathbf{k}\cdot\mathbf{g}}$$

Similarity Transformation

$$\tilde{\mathbf{H}}^{\mathbf{k}} = \mathbf{T}^{\mathbf{k}\dagger} \mathbf{H}^{\mathbf{k}} \mathbf{T}^{\mathbf{k}}$$

Diagonalization

$$\tilde{\mathbf{H}}^{\mathbf{k}} \tilde{\mathbf{A}}^{\mathbf{k}} = \tilde{\mathbf{A}}^{\mathbf{k}} \epsilon^{\mathbf{k}}$$

Back-transformation

$$\mathbf{A}^{\mathbf{k}} = \mathbf{T}^{\mathbf{k}} \tilde{\mathbf{A}}^{\mathbf{k}}$$

GPU Code

MPI process 1

Loop *on assigned $\mathbf{k}$ points*

FT

$\mathbf{H}^k \longrightarrow \text{GPU}$

$\mathbf{T}^k \longrightarrow \text{GPU}$

Similarity

Diagonalization

Back-transform

$\mathbf{e}^k \longrightarrow \text{CPU}$

$\mathbf{A}^k \longrightarrow \text{CPU}$

End Loop



GPU 1

MPI process 2

Loop *on assigned $\mathbf{k}$ points*

FT

$\mathbf{H}^k \longrightarrow \text{GPU}$

$\mathbf{T}^k \longrightarrow \text{GPU}$

Similarity

Diagonalization

Back-transform

$\mathbf{e}^k \longrightarrow \text{CPU}$

$\mathbf{A}^k \longrightarrow \text{CPU}$

End Loop

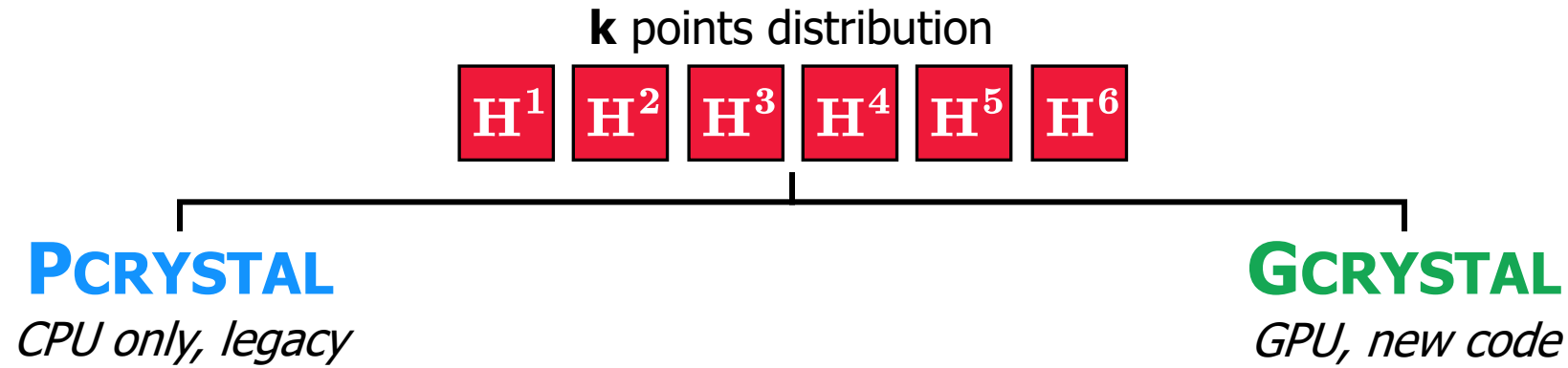


GPU 2

MPI process 3

*No assigned $\mathbf{k}$ points
Just stay idle*

Change the **k** Point Logic



Change the **k** Point Logic

k points distribution



PCRYSTAL

CPU only, legacy

GCRYSTAL

GPU, new code



Multicore CPU



One diagonalization takes 500 seconds.
All matrices are parallelized on CPU cores.

TOTAL TIME: 500 s

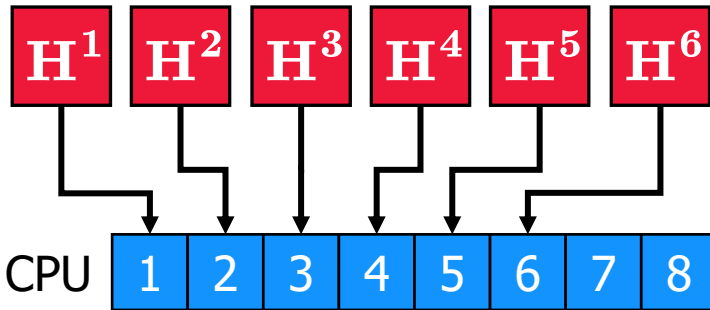
Change the **k** Point Logic

k points distribution



PCRYSTAL

CPU only, legacy



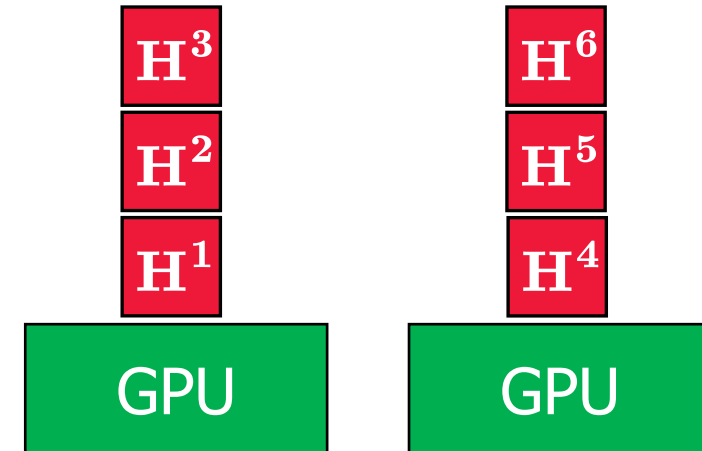
Multicore CPU

One diagonalization takes 500 seconds.
 All matrices are parallelized on CPU cores.

TOTAL TIME: 500 s

GCRYSTAL

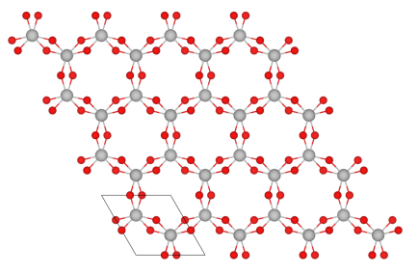
GPU, new code



One diagonalization takes 10 seconds.
 Matrices are distributed on 2 GPUs,
 operations performed in series

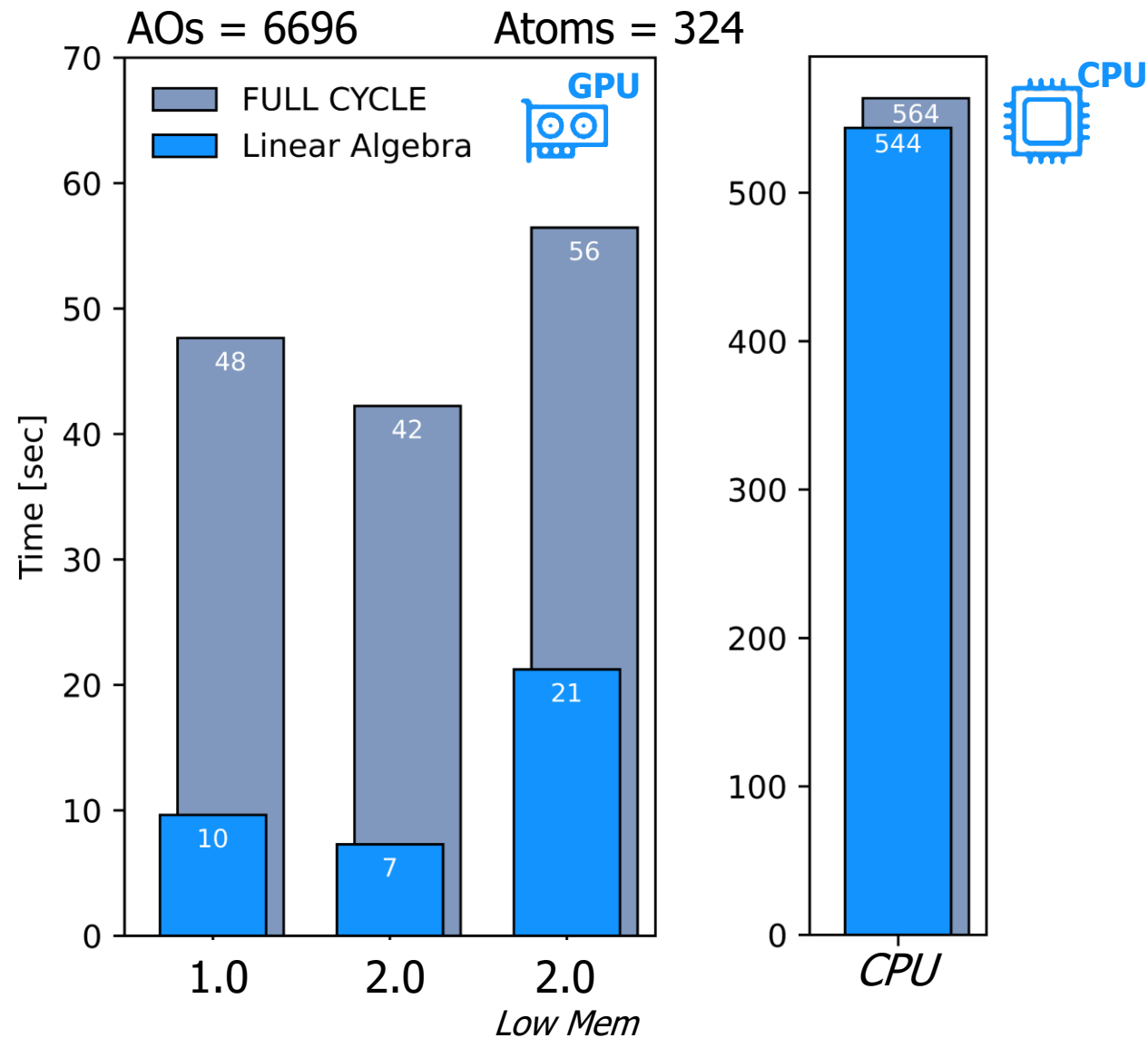
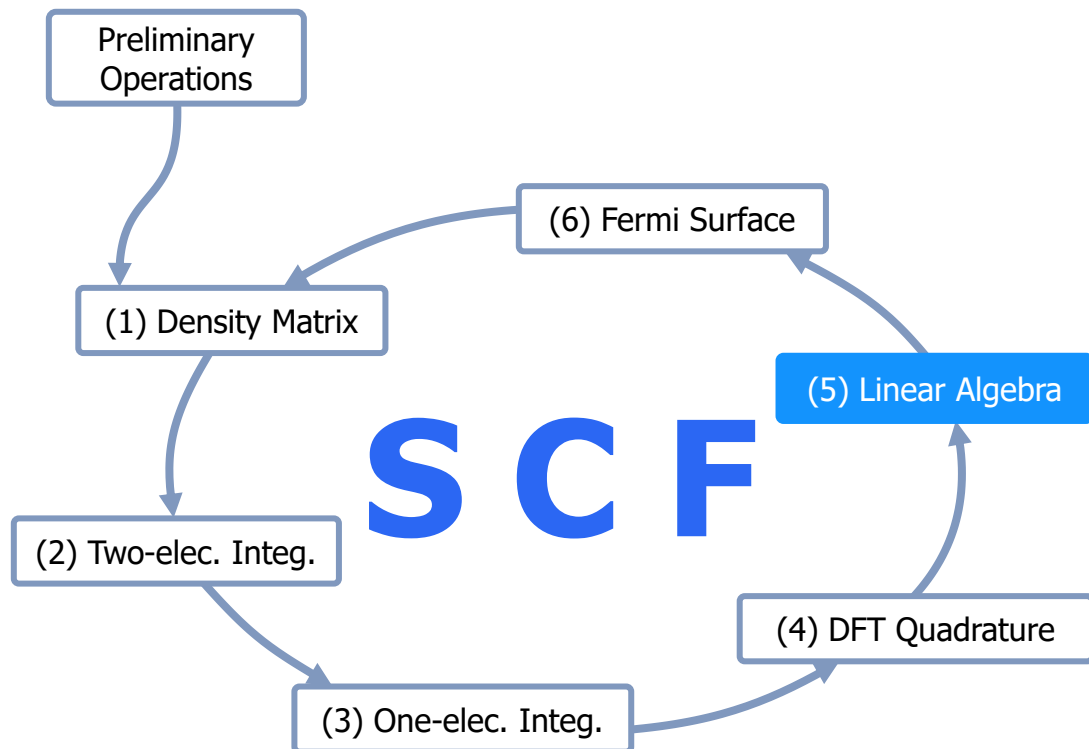
TOTAL TIME: 30 s

New GPU Development

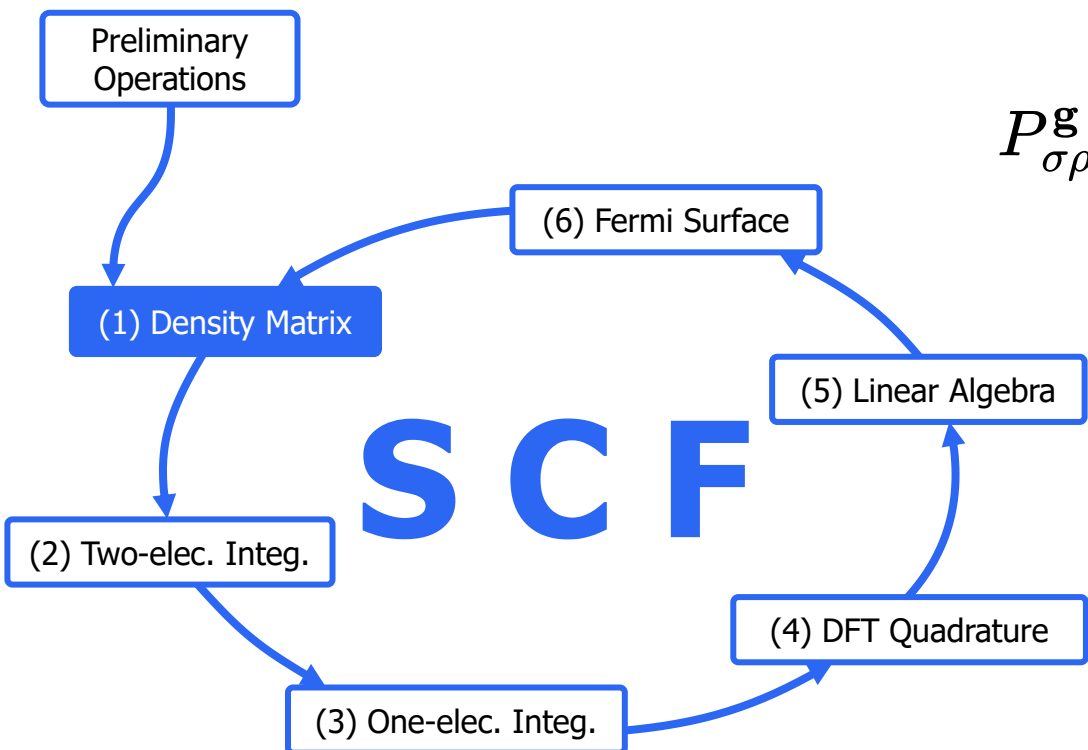


α -quartz

- PBE
- 14 **k** points
- 32 CPU cores
- 2 GPUs

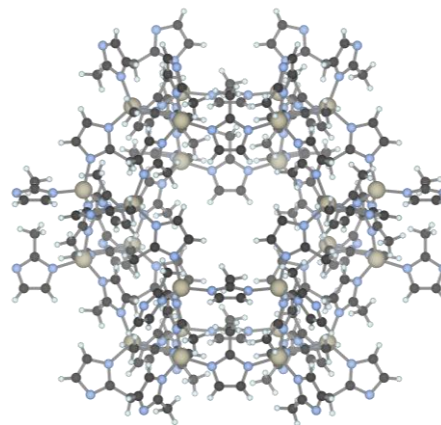


Change the **k** Point Logic



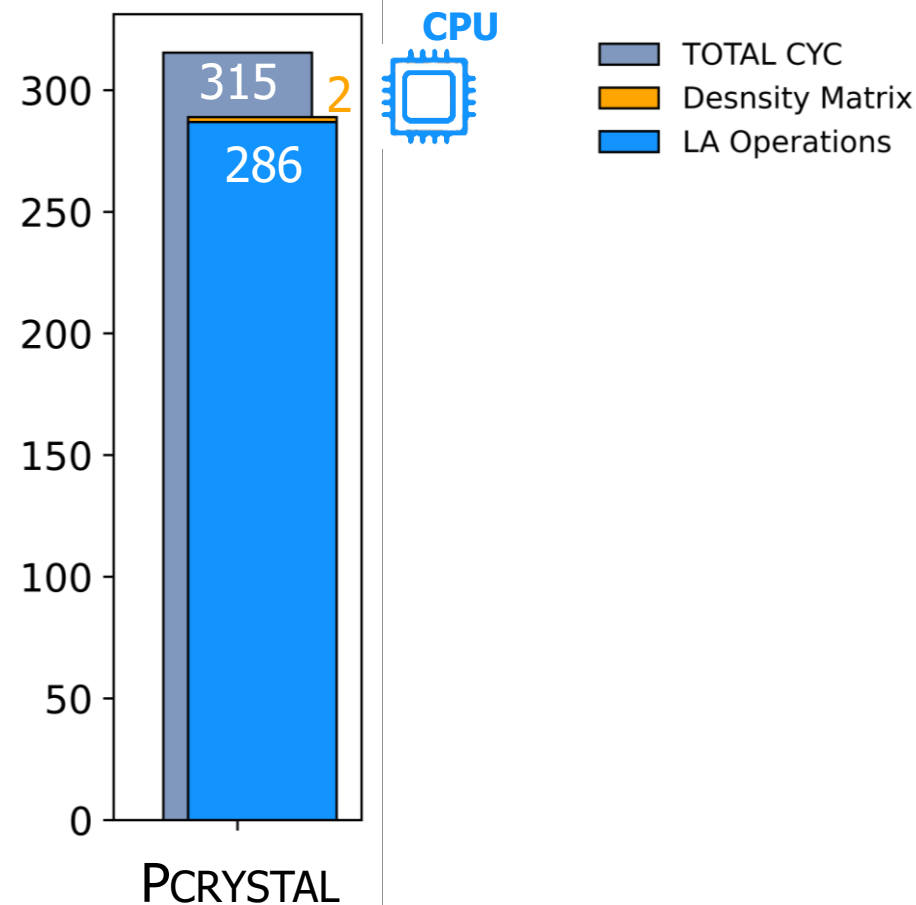
SCF

ZIF-8
276 Atoms, 2772 AOs

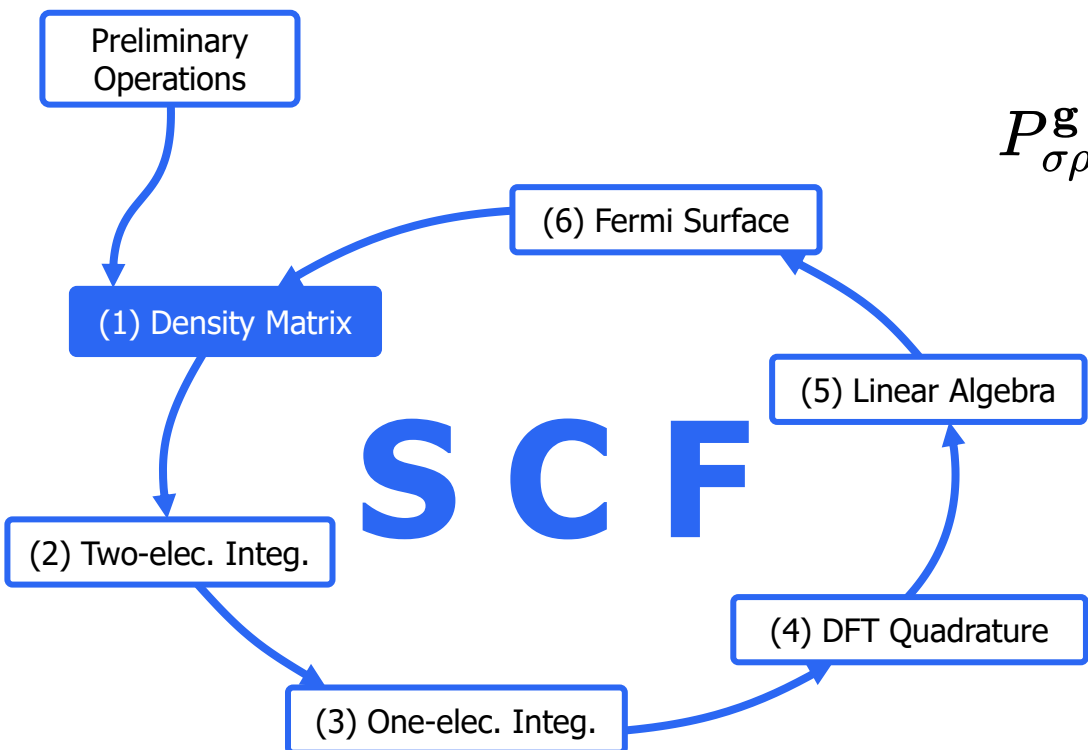


$$P_{\sigma\rho}^{\mathbf{g}} = \frac{1}{\Omega} \sum_i f_i \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{g}} a_{\sigma,i}^{\mathbf{k}} [a_{\rho,i}^{\mathbf{k}}]^* \theta[\epsilon_F - \epsilon_i^{\mathbf{k}}]$$

PCRYSTAL

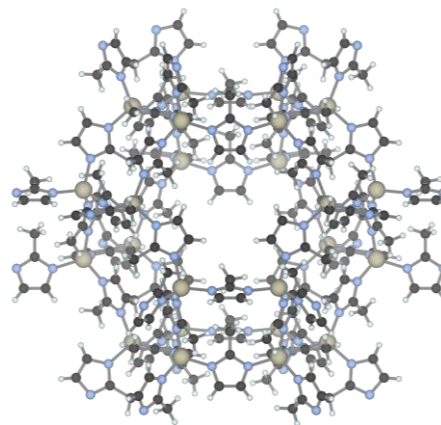


Change the **k** Point Logic



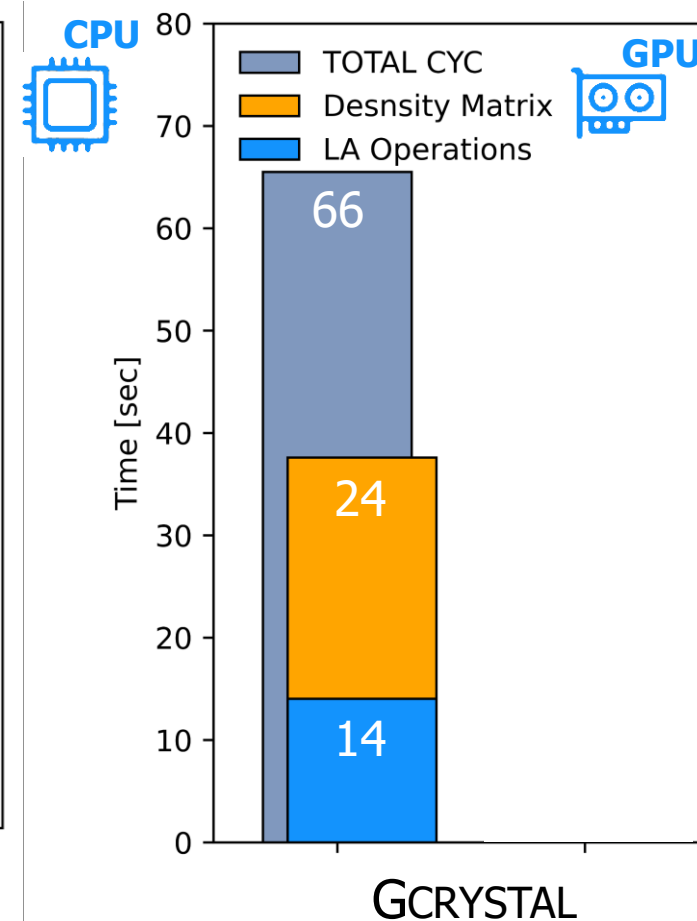
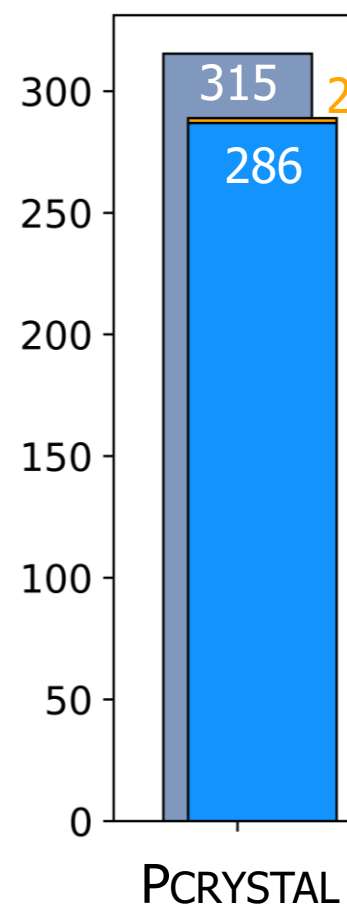
SCF

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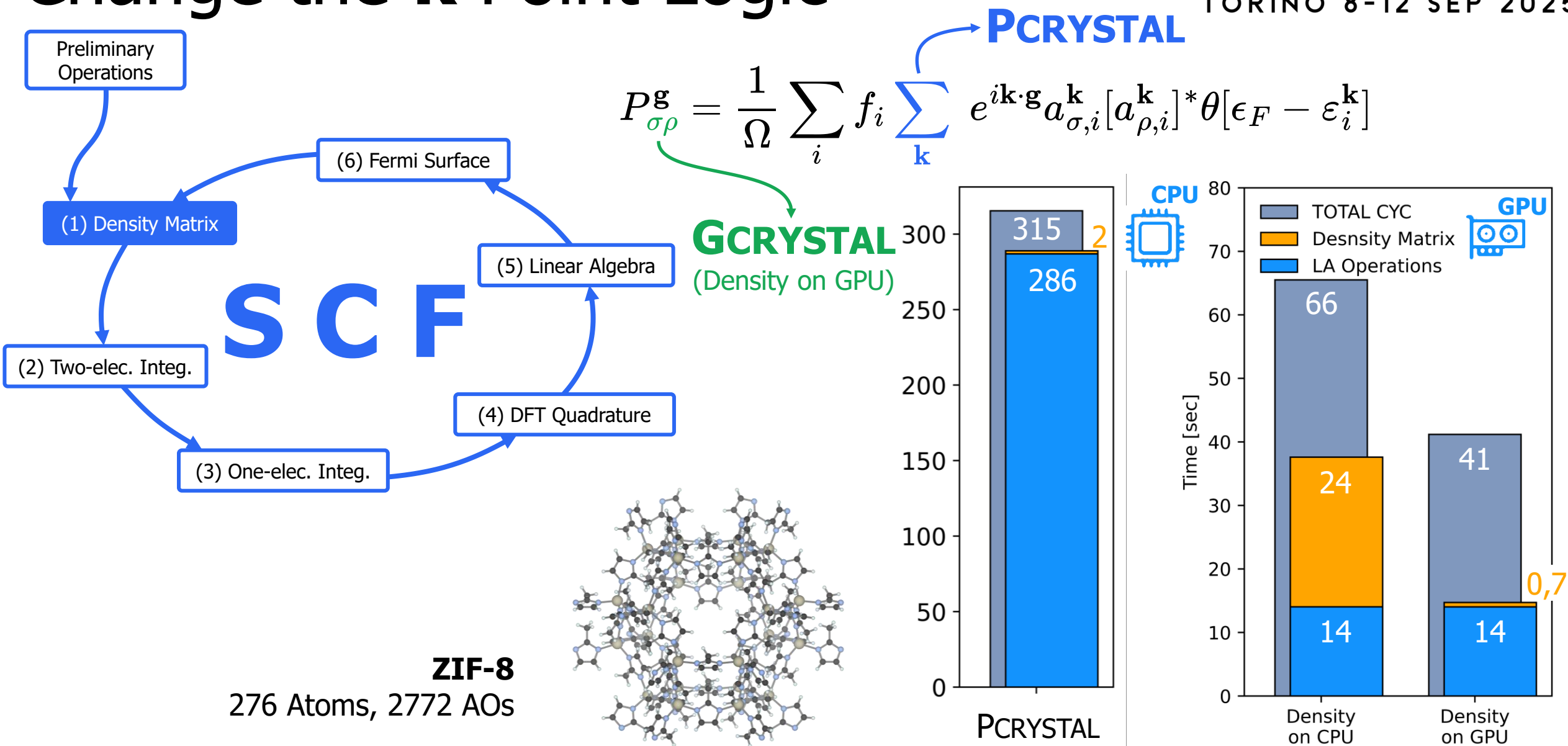


$$P_{\sigma\rho}^{\mathbf{g}} = \frac{1}{\Omega} \sum_i f_i \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{g}} a_{\sigma,i}^{\mathbf{k}} [a_{\rho,i}^{\mathbf{k}}]^* \theta[\epsilon_F - \epsilon_i^{\mathbf{k}}]$$

PCRYSTAL



Change the **k** Point Logic



Restructuring of Linear Algebra Code

- Modularity
- Exposed Parallelism
- Remove I/O (*keep only for low-mem*)
- Independence
- New **k**-Point/MPI Distribution Strategy
- Enhanced User Customization

Available for:

- **Geometry Optimization**
- **Equation-of-States**
- **Frequency Calculations**
- **Molecular Dynamics**
- **Etc.**

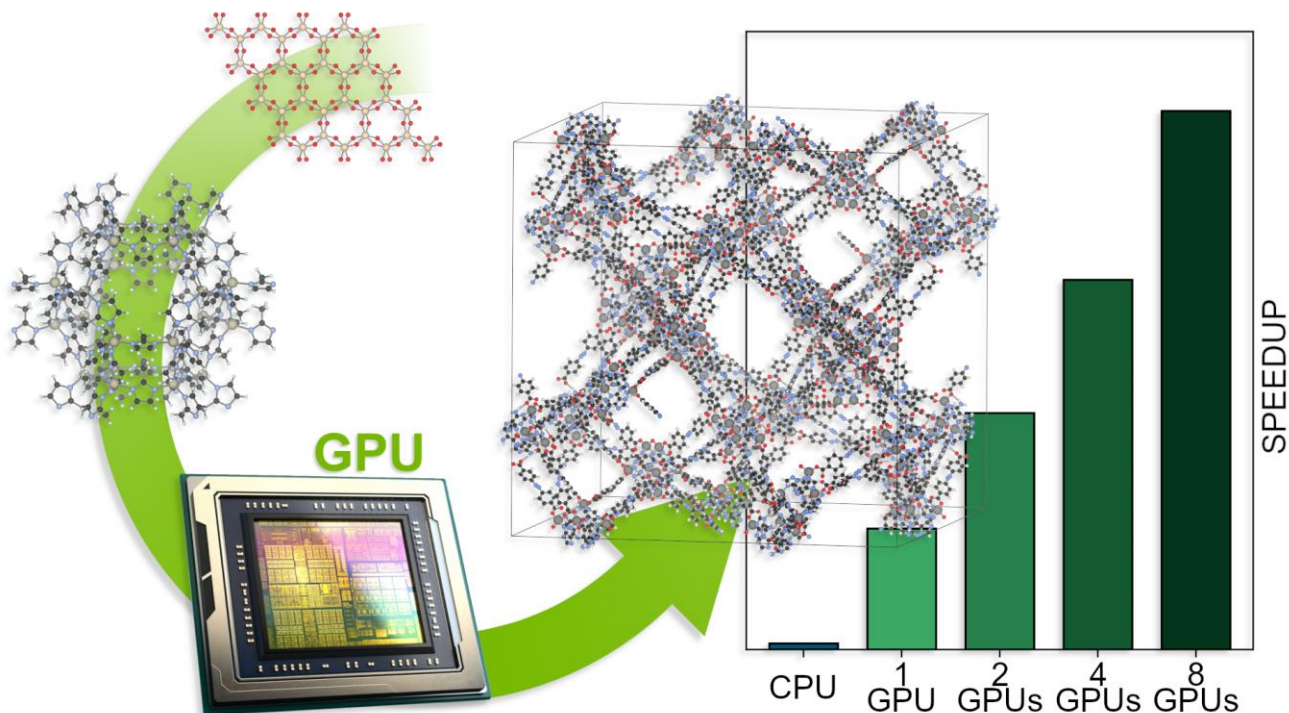
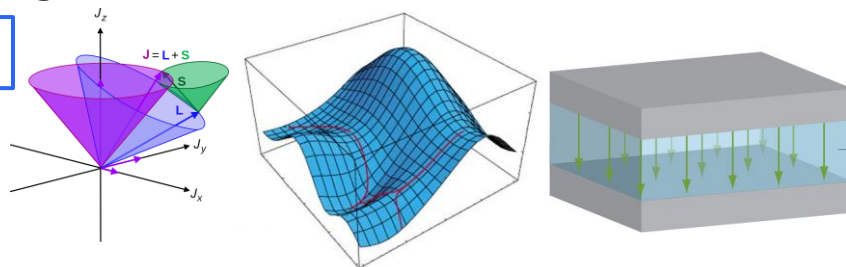
Future Perspective

Expand the GPU porting to other section of the code

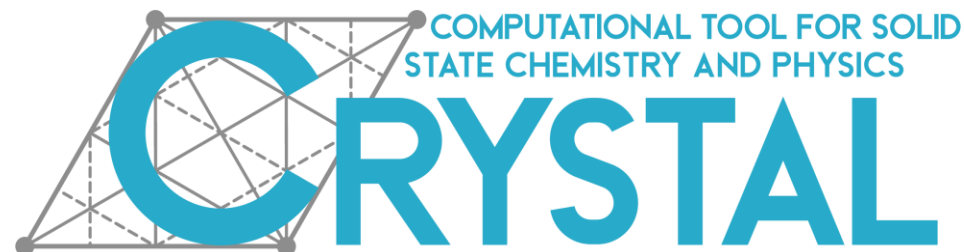
Spin-Orbit Coupling

Forces

CPHF/CPKS



GCRYSTAL will be Available in
the Next Major Release



If you are CRYSTAL users and want to try
the development build you can contact me

✉ giacomo.ambrogio@unito.it

in Giacomo Ambrogio



Acknowledgments

MSSC2025
TORINO 8-12 SEP 2025



Prof. Alessandro Erba

Prof. Bartolomeo Civalleri

Prof. Jacques Desmarais

Dr. Lorenzo Donà

Dr. Chiara Ribaldone

Dr. Filippo Spiga



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2023 - 2027
DIPARTIMENTO
DI ECCELLENZA
Ministero dell'Università e della Ricerca

Thank you for your Attention



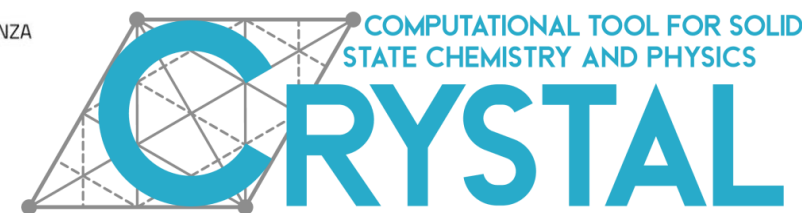
Finanziato
dall'Unione europea
NextGenerationEU



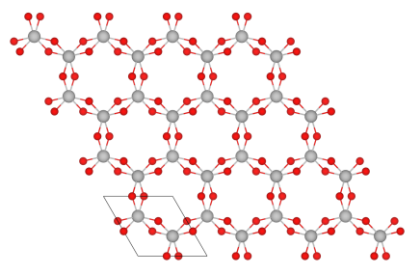
Italiadomani
PIANO NAZIONALE DI RIPRESA E RESILIENZA



EuroHPC
Joint Undertaking



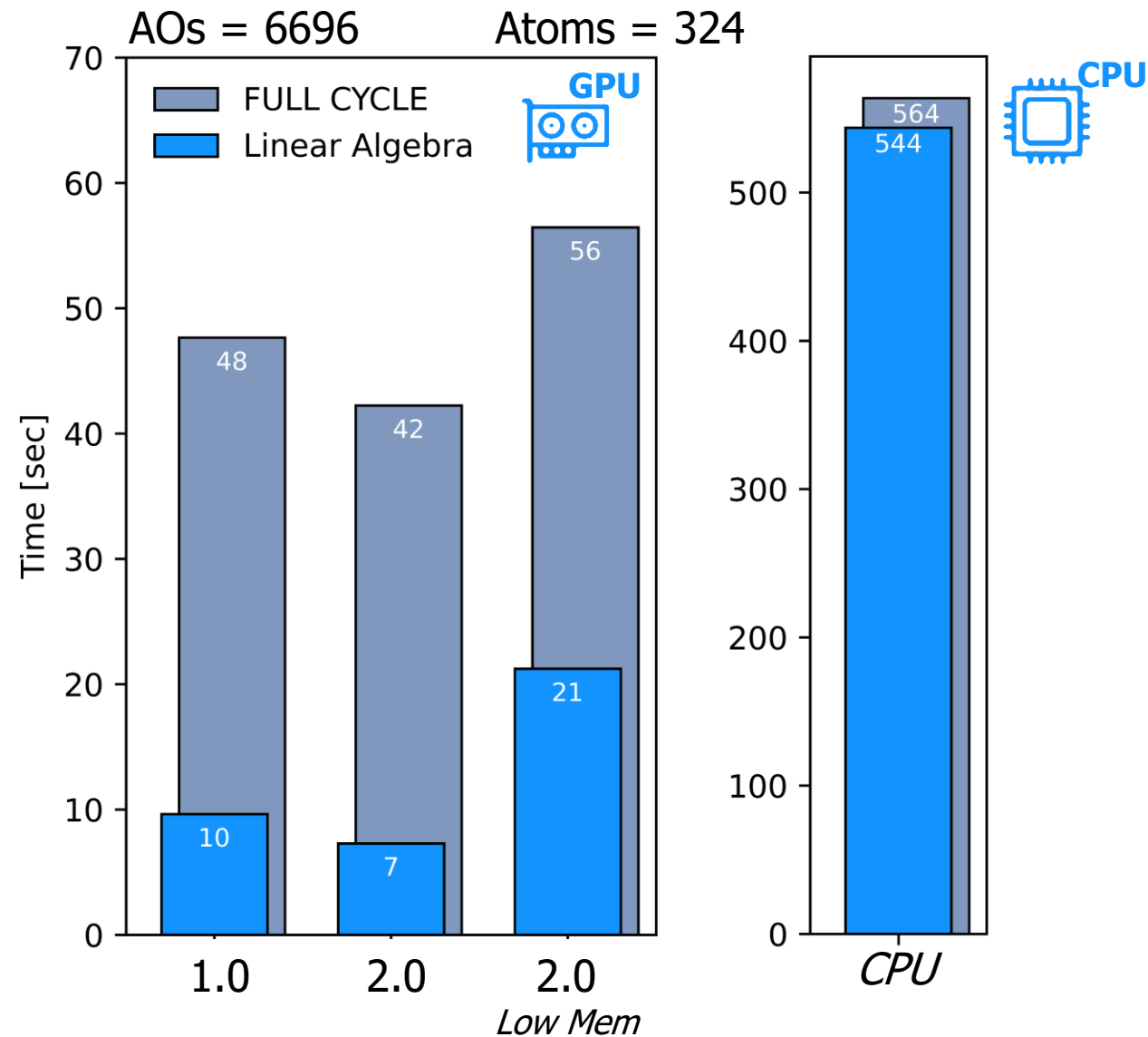
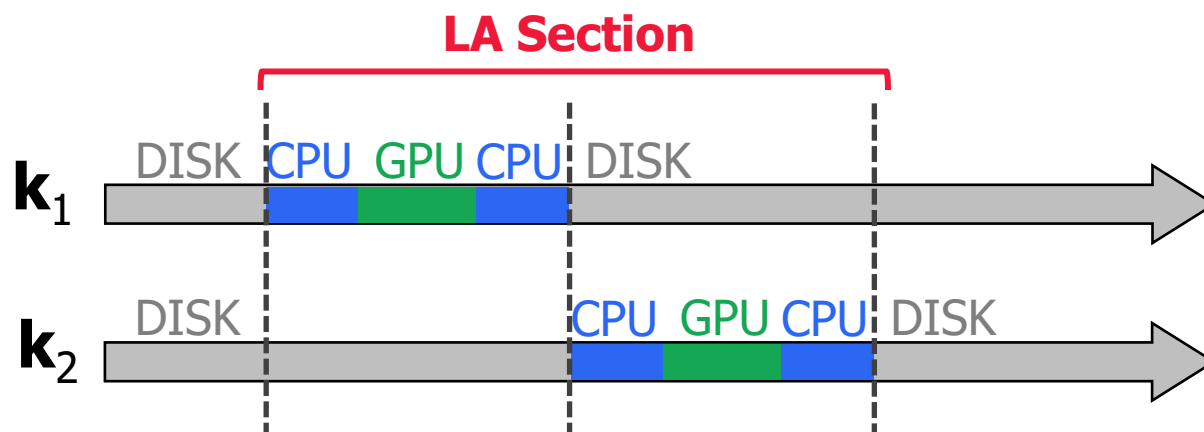
New GPU Development



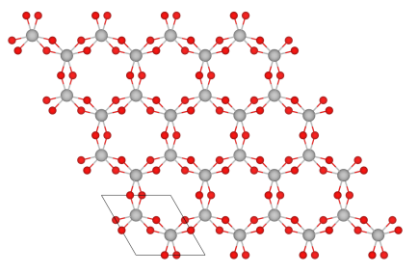
α -quartz

- PBE
- 14 **k** points
- 32 CPU cores
- 2 GPUs

GCRYSTAL 1.0



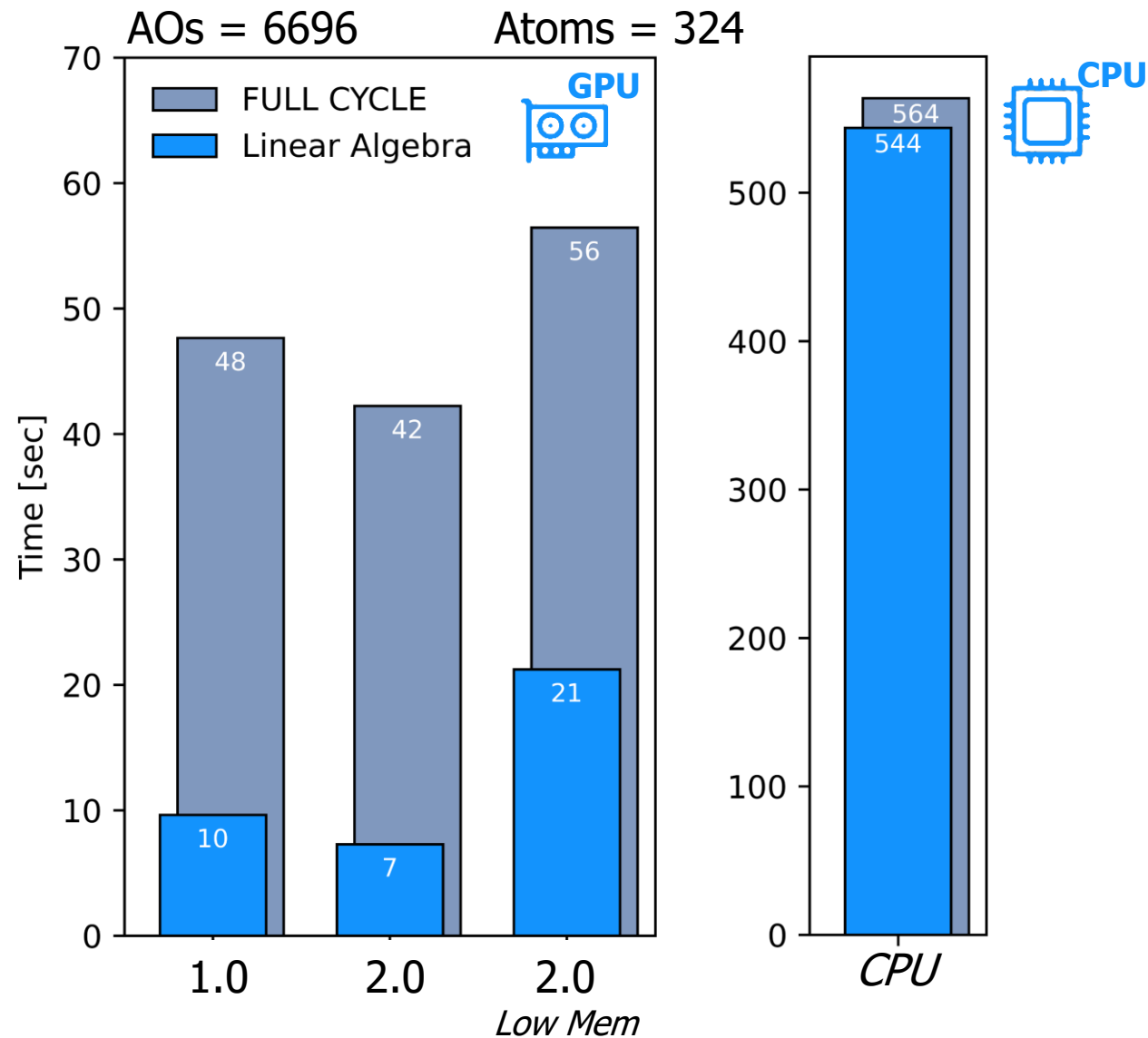
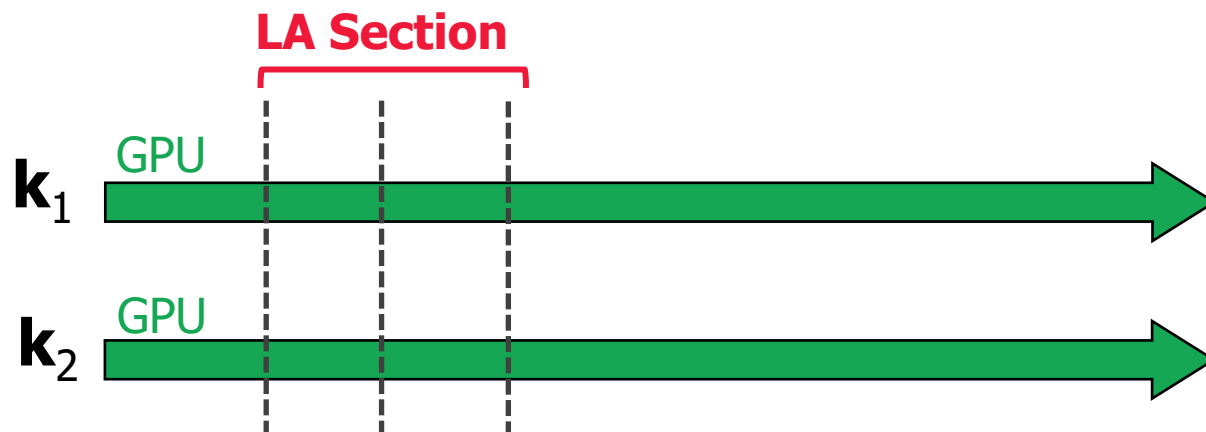
New GPU Development



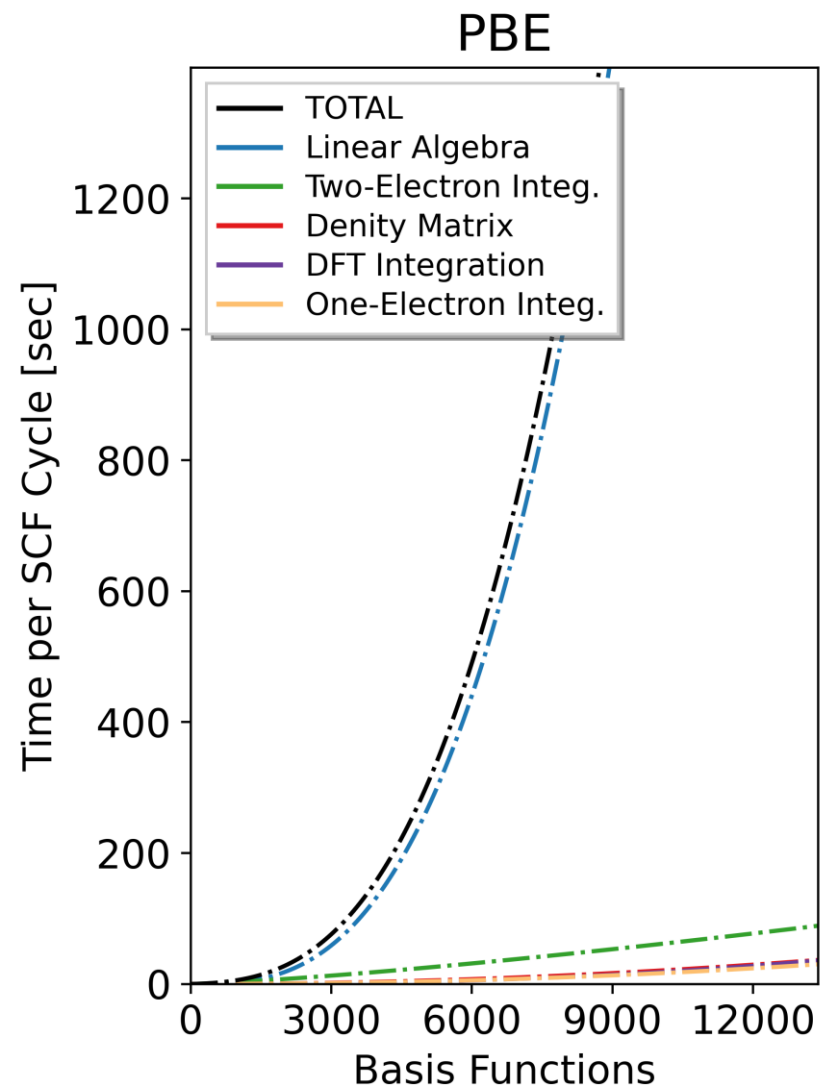
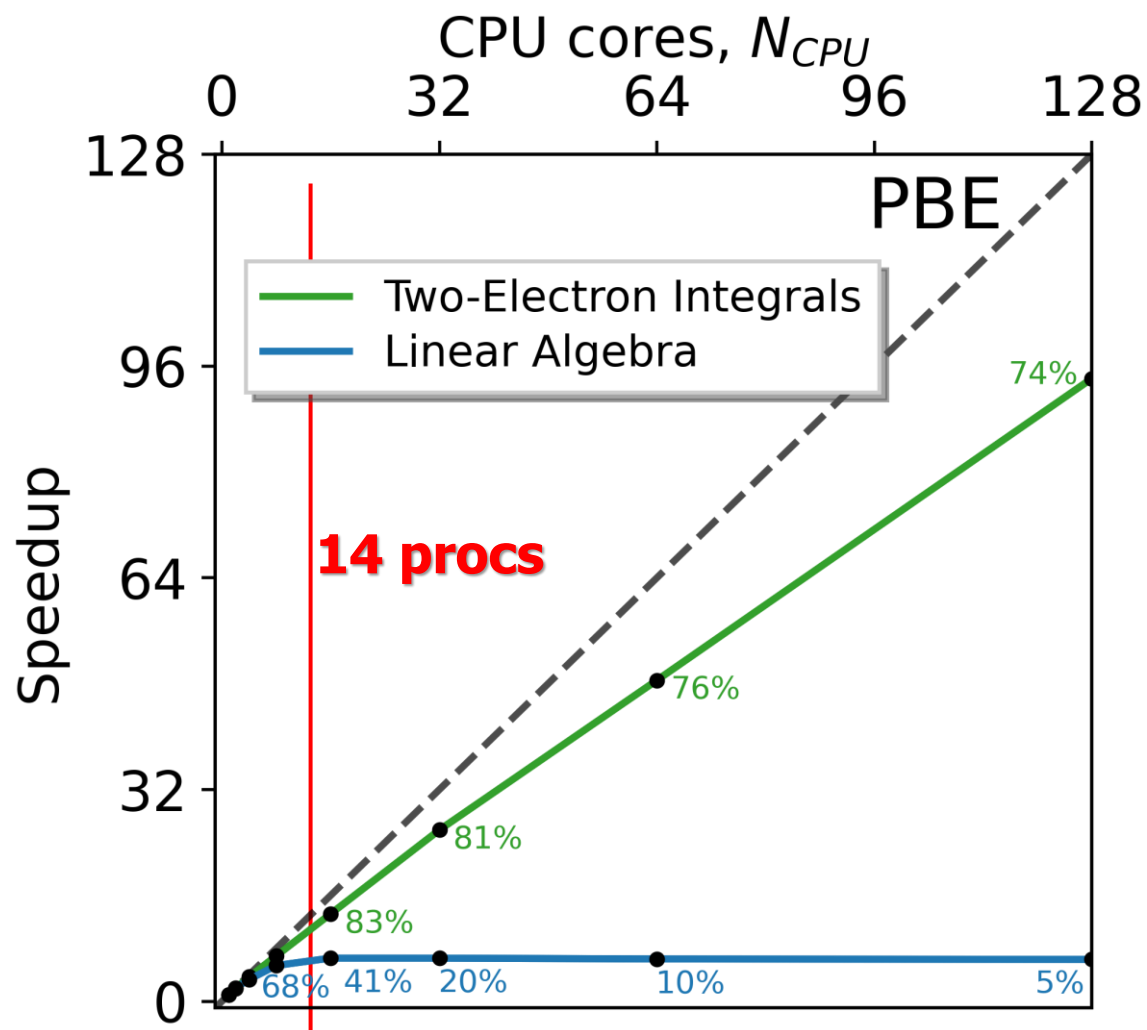
α -quartz

- PBE
- 14 **k** points
- 32 CPU cores
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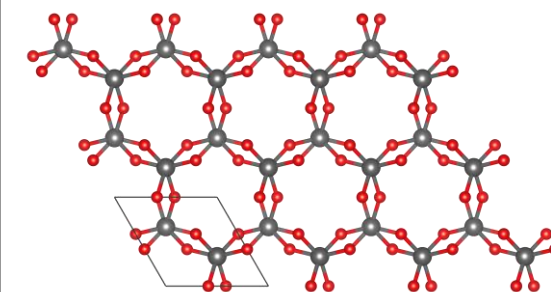
GCRYSTAL 2.0



Scaling in CPU PCRYSTAL

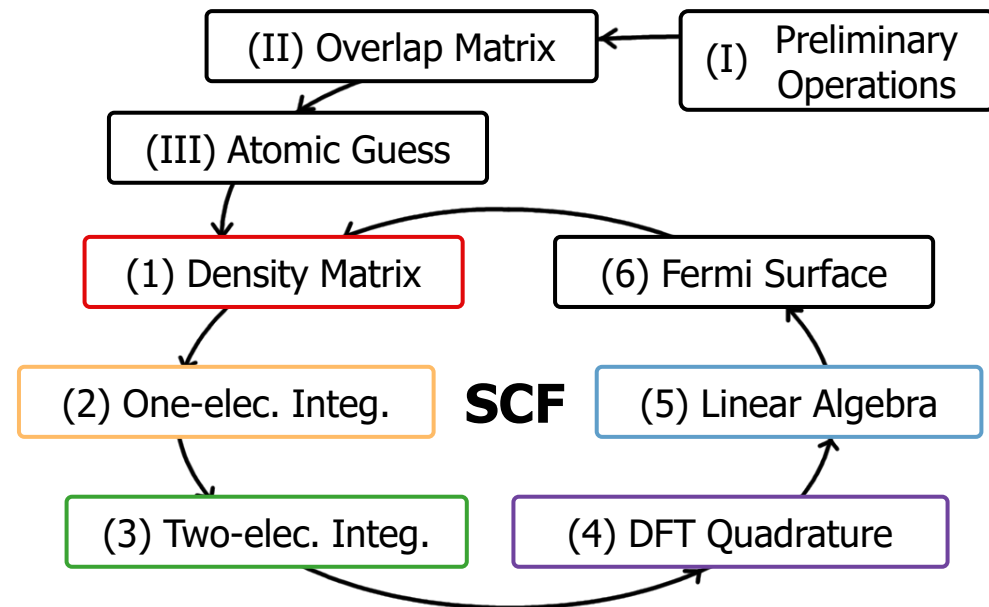


α -quartz



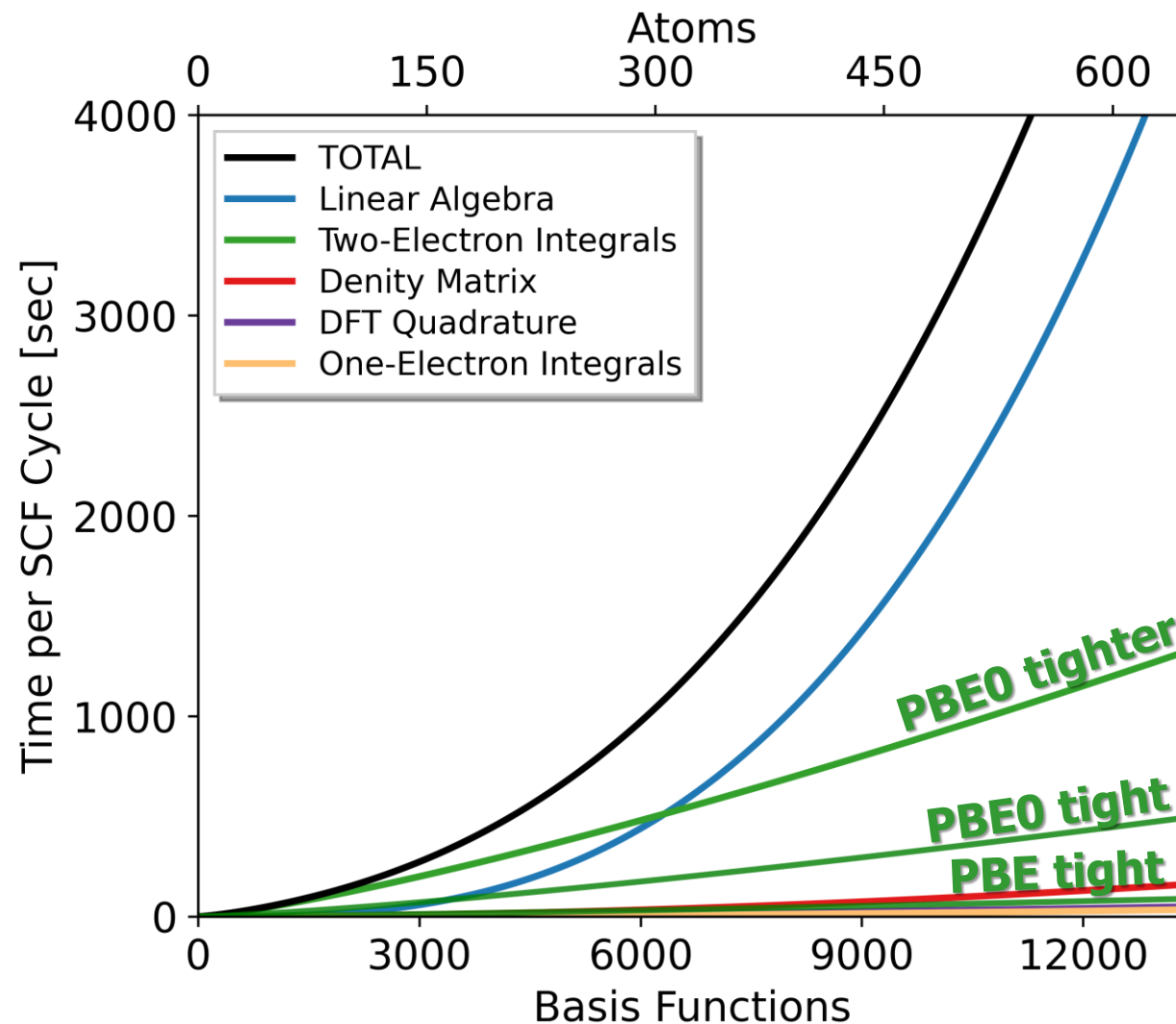
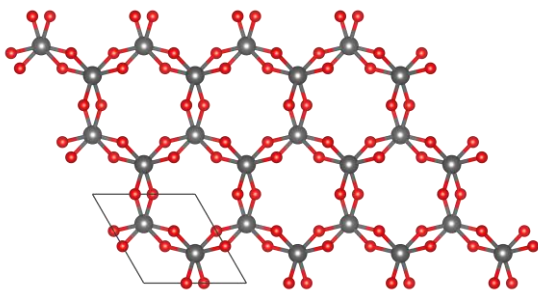
- PBE
- Tight Tolerances
- 14 $\mathbf{k}$ Points
- 1-128 procs

Scaling in CPU PCRYSTAL



α -quartz

- PBE
- Tight Tolerances
- 14 **k** Points
- 1-128 procs



SCF Steps in CRYSTAL

(2) Two-elec. Integ.

$$E_{\text{coul}} = \frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\nu}^{\mathbf{g}} | \chi_{\sigma}^{\mathbf{s}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

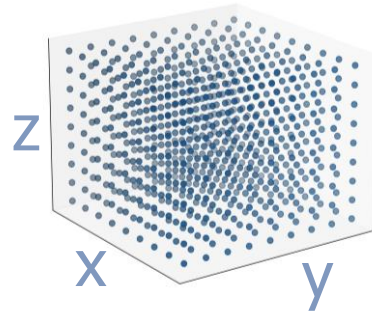
(3) One-elec. Integ.

$$E_{\text{kin}} = \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \langle \chi_{\mu}^0 | -\frac{\nabla^2}{2} | \chi_{\nu}^{\mathbf{g}} \rangle$$

$$E_{\text{e-nucl}} = -\sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\lambda} Z_{\lambda} \sum_{\mathbf{h}} \left\langle \chi_{\mu}^0 \left| \frac{1}{|\mathbf{r} - \mathbf{h} - \mathbf{s}_{\lambda}|} \right| \chi_{\nu}^{\mathbf{g}} \right\rangle$$

(4) DFT Quadrature

$$E_{\text{xc}}[\rho(\mathbf{r})] = \int d\mathbf{r} \varepsilon_{\text{c}}[\rho(\mathbf{r})] + \int d\mathbf{r} \varepsilon_{\text{x}}[\rho(\mathbf{r})]$$



(6) Fermi Surface

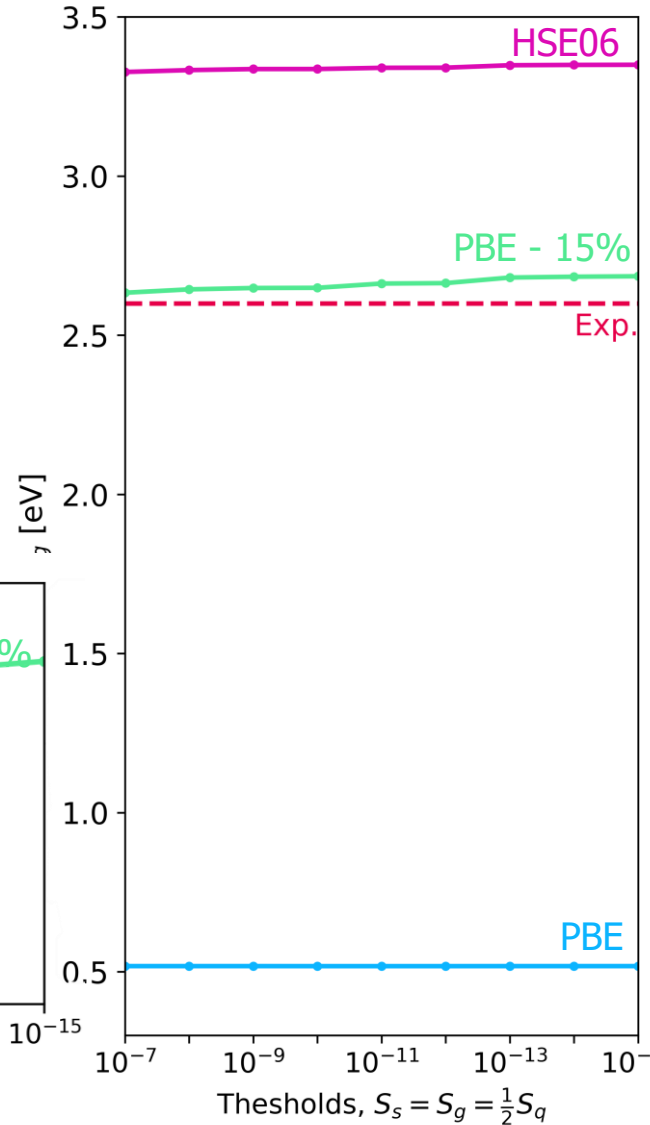
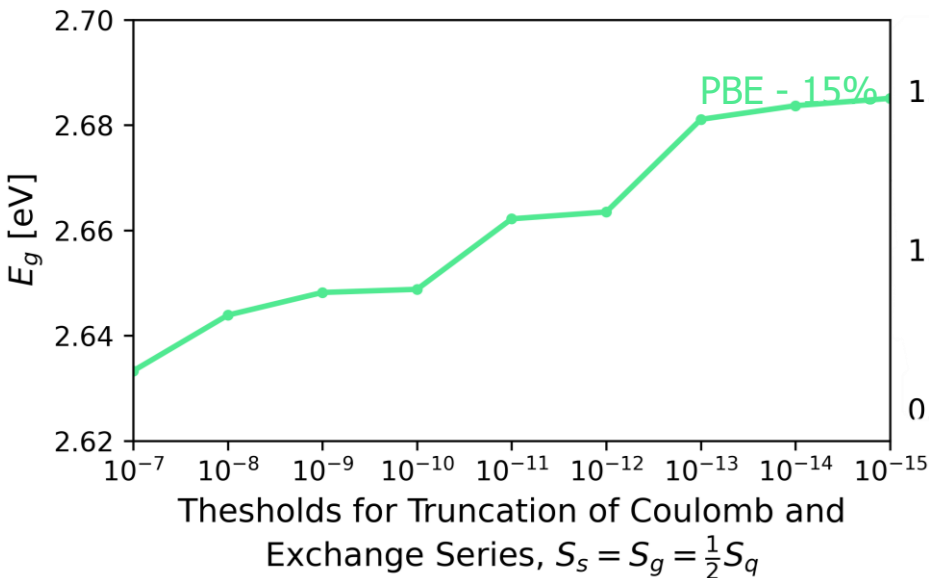
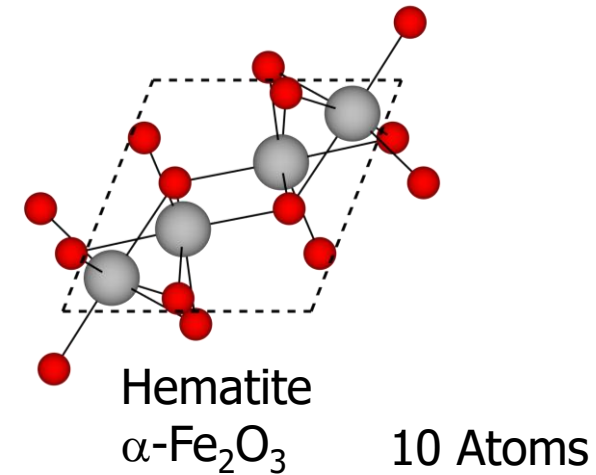
$$n \text{ elect.} = \frac{1}{\Omega} \int_{\text{FBZ}} d\mathbf{k} \sum_i \theta[\epsilon_F - \epsilon_i^{\mathbf{k}}] \quad \text{Not parallelized}$$

(1) Density Matrix

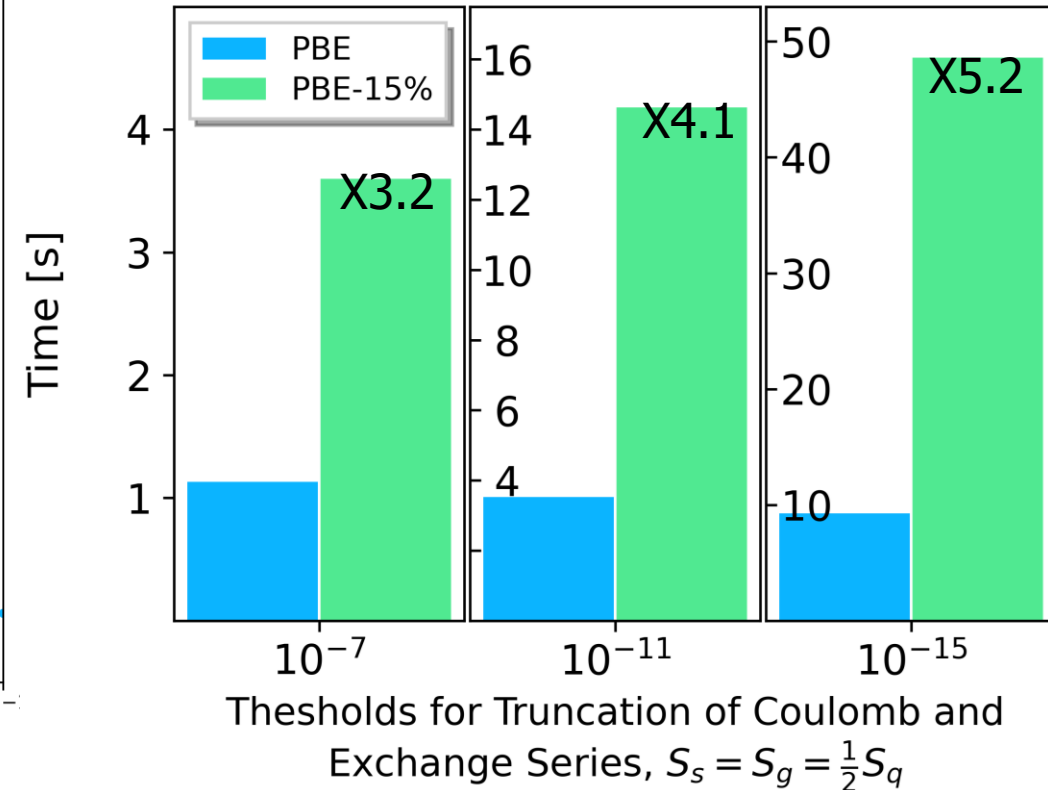
$$P_{\sigma\rho}^{\mathbf{g}} = \frac{1}{\Omega} \sum_i f_i \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{g}} a_{\sigma,i}^{\mathbf{k}} [a_{\rho,i}^{\mathbf{k}}]^* \theta[\epsilon_F - \epsilon_i^{\mathbf{k}}]$$

Parallelized over $\mathbf{k}$ points

Evolution of Thresholds



Computation of Two-electron Integrals (Coulomb + EXX)



More Preliminary Tests

	k Points	AOs	Calculation Type	Total Timings		LA SPEEDUP	TOT SPEEDUP
				GPU [sec] 	CPU [sec] 		
Quartz (<i>old code</i>)	28	248	Single Point	7.56	3.45	0.27	0.46
Quartz	28	248	Single Point	4.17	3.45	0.42	0.83
KMnF ₃	24 - 72	702	Phonon	442	423	2.15	0.96
Andradite	6	1724	EOS	4765	4820	8.00	1.01
128 H ₂ O box	14	3072	MD (4 steps)	2958	7331	37.8	2.48
Quartz 3x3x3	7	6696	Single Point	403	5127	73.5	12.7

Local Cluster

- 2x AMD 9124 EPYC 16 cores CPUs
- 2x NVIDIA A100 GPUs (80GB)
- 384 GB Memory

**Density Constructed
on CPU!**

Test Case: Giant MOF

MSSC2025
TORINO 8-12 SEP 2025

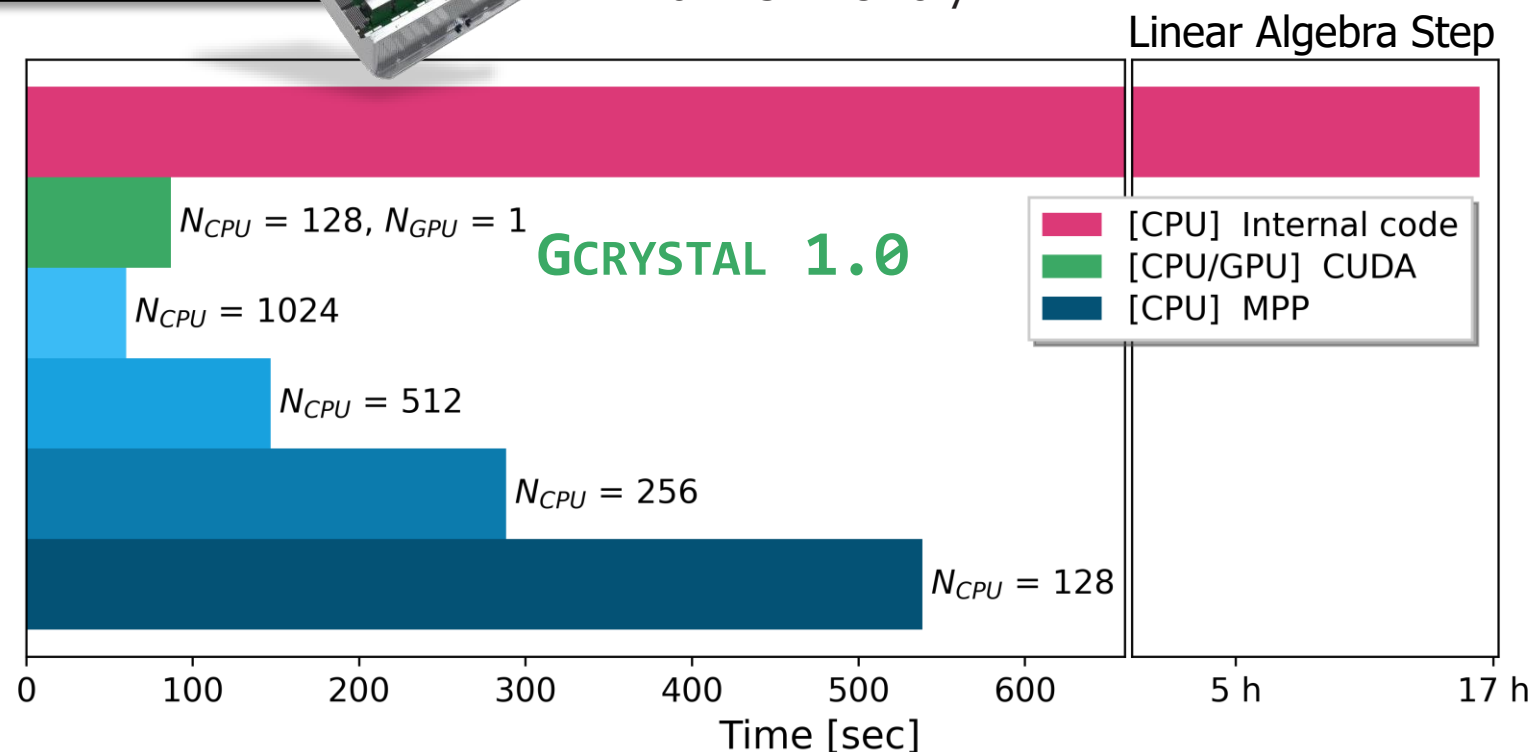
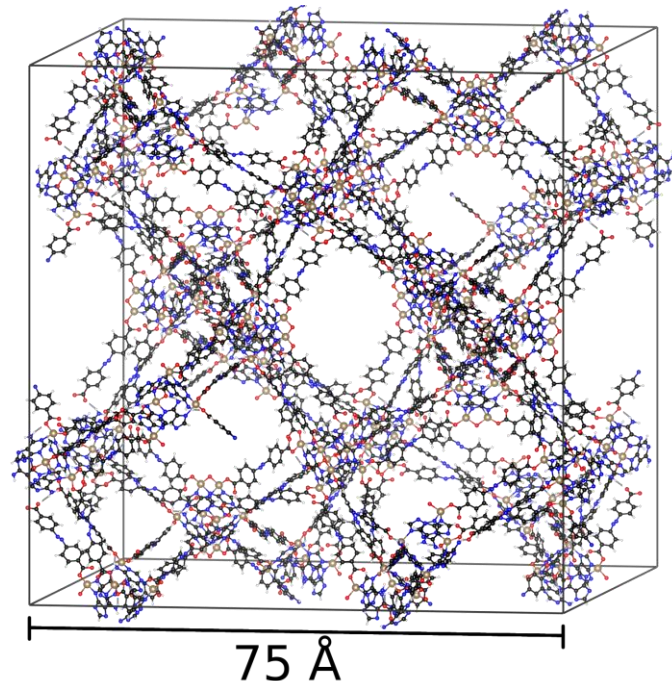
bio-MOF-102

- 2808 Atoms/Cell
- 33672 Basis Functions
- PBEsol0
- 1 **k** Point



GPU Partition

- 2x AMD 64 cores
- 8x NVIDIA A100 (40GB)
- 1024 GB Memory



G. Ambrogio, et al. "Accelerated linear algebra for large scale DFT calculations of materials on CPU/GPU architectures with CRYSTAL." *The Journal of Chemical Physics* 162.8 (2025).